RD Sharma Class X Mathematics

Board-Important & HOTS Questions (2026–27)

Filtered from RD Sharma Mathematics – Class X (2025–26 edition)

Curated using the book's own BASIC/LOTS/HOTS tiers and CBSE/NCERT exam tags

How this book was filtered: RD Sharma tags many of its own exercise questions with their difficulty tier (**BASIC**, **LOTS** = Lower Order Thinking Skills, **HOTS** = Higher Order Thinking Skills) and, for many questions, the exact board exam they are drawn from (e.g. *[CBSE 2024]*) or NCERT source. This compilation keeps: every LOTS and HOTS question; every CBSE- or NCERT-tagged question regardless of tier; and a representative sample of untagged BASIC questions for foundational coverage. Source tags are shown in *gold* beside each question. Answers are reproduced from the book's own Answer/Hint keys.
Syllabus alignment: per the CBSE 2026–27 syllabus, the following have been excluded throughout: Euclid's Division Lemma/Algorithm; decimal expansion (terminating/non-terminating); the cubic-polynomial zero–coefficient relation and the Division Algorithm for polynomials; the Cross-Multiplication Method and equations reducible to a pair of linear equations; equations reducible to quadratic form and the method of completing the square; the coordinate-geometry Area-of-a-Triangle formula; formal proofs of the Basic Proportionality Theorem, Pythagoras Theorem, and circle-tangent theorems (applications are retained); trigonometric ratios of complementary angles; ogives (graphical cumulative frequency); and the frustum of a cone. Constructions does not appear as a chapter in this edition.

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CHAPTER 1

Real Numbers

*Filtered from RD Sharma Class X (2025–26 edition). This chapter's own difficulty tiers — **BASIC**, **LOTS** (Lower Order Thinking Skills) and **HOTS** (Higher Order Thinking Skills) — are used as the primary filter: all LOTS and HOTS items are retained, along with CBSE/NCERT-tagged items from the BASIC tier and a representative sample of the remainder. Per the CBSE 2026–27 syllabus, **Euclid's Division Lemma/Algorithm** and the **terminating/non-terminating decimal expansion** topic have been removed and are excluded below.*

Key Concepts

- **Fundamental Theorem of Arithmetic:** every composite number can be expressed as a product of primes, uniquely, apart from the order of factors.
- **HCF/LCM by prime factorisation:** HCF = product of common prime factors raised to their *smallest* exponents; LCM = product of all prime factors (from either number) raised to their *greatest* exponents.
- For any two positive integers a, b: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$ (this relation does *not* extend to three or more integers).
- $\sqrt{2}, \sqrt{3}, \sqrt{5}$, and sums/differences of a nonzero rational with an irrational, are irrational; standard proofs use contradiction via the Fundamental Theorem of Arithmetic.

Exercise 1.1 — Divisibility & Prime Factorisation

1. Express each of the following as a product of its prime factors: (i) 420 (ii) 468
2. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.
3. Explain why $3 \times 5 \times 7 + 7$ is a composite number. *[NCERT Exemplar]*
4. Check whether 6^n can end with the digit 0 for any natural number n. *[NCERT; CBSE 2023]*
5. Can the number $(15)^n$, n being a natural number, end with the digit 0? Give reasons. *[CBSE 2024]*
6. Find the values of x, y, z, w in the factor tree below (top to bottom: x splits into 2 and a node 819; 819 splits into 3 and y; y splits into 3 and z; z splits into 7 and w; w splits into 7 and 13). Also write the prime factorisation of x. *[CBSE 2024]*
7. If p is a prime and p divides a^2, prove that p divides a, for a positive integer a. *[NCERT; HOTS]*

Answers: 1. (i) $2^2 \times 3 \times 5 \times 7$ (ii) $2^2 \times 3^2 \times 13$. 2. $7 \times 11 \times 13 + 13 = (7 \times 11 + 1) \times 13$; $7\times6\times5\times4\times3\times2\times1+5 = (7\times6\times4\times3\times2\times1+1)\times5$. 3. $3\times5\times7+7 = (15+1)\times7 = 16\times7$. 4. No. 5. No. 6. $x = 3276$, $y = 1639$, $z = 273$, $w = 91$; $x = 2^2 \times 3^2 \times 7 \times 13$. 7. Proved via Fundamental Theorem of Arithmetic (prime factorisation argument).

Exercise 1.2 — HCF, LCM and Applications

1. Find the LCM and HCF of 96 and 120, and verify that LCM $\times$ HCF = product of the numbers.

2. Find the LCM and HCF of the following by prime factorisation: (i) 12, 15 and 21 *[NCERT]* (ii) 8, 9 and 25 *[NCERT]*

3. Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$. *[NCERT]*

4. Can two numbers have 16 as their HCF and 380 as their LCM? Give reasons.

5. Three bells ring at intervals of 6, 12 and 18 minutes. If all three rang together at 6 a.m., when will they next ring together? *[CBSE 2023, 2024]*

6. Find the smallest number which, when increased by 17, is exactly divisible by both 520 and 468.

7. Find the largest number which, on dividing 1251, 9377 and 15628, leaves remainders 1, 2 and 3 respectively. *[CBSE 2019]*

8. On a morning walk, three persons step off together; their steps measure 30 cm, 36 cm and 40 cm. What is the minimum distance each should walk so that each covers the same distance in complete steps? *[CBSE 2019]*

9. The traffic lights at three road crossings change after every 48, 72 and 108 seconds. If they change simultaneously at 7 a.m., when will they next change together? *[CBSE 2023]*

10. Find the smallest number which leaves remainders 8 and 12 when divided by 28 and 32, respectively. *[HOTS]*

11. Find the greatest number of 6 digits exactly divisible by 24, 15 and 36. *[HOTS]*

12. Prove that $\sqrt{3}+\sqrt{5}$ is irrational, using the Fundamental Theorem of Arithmetic based argument (not Euclid's Algorithm). *[NCERT Exemplar; HOTS]*

Answers: 1. HCF = 24, LCM = 480. 2. (i) LCM = 420, HCF = 3 (ii) LCM = 1800, HCF = 1. 3. 22338. 4. No. 5. 6:36 a.m. 6. 4663. 7. 625. 8. 360 cm. 9. 7:07:12 a.m. 10. 204. 11. 999720. 12. Proved by contradiction.

Exercise 1.3 — Irrationality of Numbers

1. Show that $\dfrac{1}{\sqrt{2}}$ and $6 + \sqrt{2}$ are irrational.

2. Show that $3 + \sqrt{2}$ is an irrational number. *[CBSE 2009]*

3. Prove that $4 - 5\sqrt{2}$ is an irrational number. *[CBSE 2010]*

4. Prove that $2\sqrt{3} - 1$ is an irrational number. *[CBSE 2010]*

5. Prove that $2 - 3\sqrt{5}$ is an irrational number. *[CBSE 2010]*

6. Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number. *[CBSE 2018]*

7. Prove that $\dfrac{2+\sqrt{3}}{5}$ is an irrational number, given that $\sqrt{3}$ is irrational. *[CBSE 2019]*

8. Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is irrational. *[CBSE 2019]*

9. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. *[NCERT Exemplar]*

10. Prove that $2 + \sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. *[CBSE 2023]*

11. Prove that $5 - 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. *[CBSE 2024]*

12. Prove that $\dfrac{2-\sqrt{3}}{5}$ is an irrational number, given that $\sqrt{3}$ is irrational. *[CBSE 2024]*

13. Prove that $(2 + \sqrt{6})$ is an irrational number, given that $\sqrt{6}$ is irrational. *[CBSE 2024]*

14. Prove that for any prime positive integer p, $\sqrt{p}$ is irrational. *[HOTS]*

15. If p, q are prime positive integers, prove that $\sqrt{p} + \sqrt{q}$ is an irrational number. *[NCERT Exemplar; HOTS]*

(All items above are proof-based; methods follow the standard contradiction argument used with the Fundamental Theorem of Arithmetic — see Key Concepts.)

Very Short Answer Questions (VSAQs)

1. What is the total number of factors of a prime number? *[CBSE 2020]*

2. State the Fundamental Theorem of Arithmetic.

3. Write 98 as a product of its prime factors.

4. If the prime factorisation of a natural number n is $2^3 \times 3^2 \times 5^2 \times 7$, write the number of consecutive zeros in n.

5. If the product of two numbers is 1080 and their HCF is 30, find their LCM.

6. If p and q are two prime numbers, what is their HCF? Their LCM?

7. What is the HCF of the smallest composite number and the smallest prime number? *[CBSE 2018]*

8. HCF of two numbers is always a factor of their LCM (True/False).

9. Express 429 as a product of its prime factors. *[CBSE 2019]*

10. Two positive integers a and b can be written as $a = x^3y^2$ and $b = xy^3$, where x, y are primes. Find LCM(a, b). *[CBSE 2019]*

11. If $\text{HCF}(336, 54) = 6$, find $\text{LCM}(336, 54)$. *[CBSE 2019]*

12. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. *[CBSE 2024]*

13. If a and b are relatively prime numbers, what is their HCF? Their LCM?

Answers: 1. 2. 2. See key concepts. 3. 2×7^2. 4. 2. 5. 36. 6. HCF = 1, LCM = pq. 7. 2. 8. True. 9. $3 \times 11 \times 13$. 10. x^3y^3. 11. 3024. 12. Composite $(= 11 \times (5 \times 17 + 3) = 11 \times 88)$. 13. HCF = 1, LCM = ab.

Fill in the Blanks

1. If $\text{HCF}(306, 657) = 9$, then $\text{LCM}(306, 657) =$________.

2. If a and b are two positive co-prime integers such that $a = 12b$, then $\text{HCF}(a, 12) =$________.

3. If two positive integers m, n are expressible as $m = a^2b^3$ and $n = a^3b^2$ (a, b prime), then $\text{HCF}(m, n) =$________ and $\text{LCM}(m, n) =$________.

4. The ratio between the HCF and LCM of 5, 15 and 20 is ________.

5. Two numbers are in the ratio 21 : 17. If their HCF is 5, the numbers are ________ and ________.

6. Two numbers are in the ratio 3 : 4 and their LCM is 120. The sum of the numbers is ________.

7. The LCM of the smallest prime number and the smallest composite number is ________. *[HOTS]*

Answers: 1. 22338. 2. 12. 3. HCF = a^2b^2, LCM = a^3b^3. 4. 1 : 12. 5. 85, 105. 6. 70. 7. 4.

CHAPTER 2

Polynomials

Per the CBSE 2026–27 syllabus, the relation between zeros and coefficients of ***cubic*** *polynomials, and the* ***Division Algorithm*** *for polynomials, have been removed; only quadratic-polynomial content is retained below.*

Key Concepts

- If α, β are the zeros of $ax^2 + bx + c$: $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$.
- A quadratic polynomial with sum S and product P of zeros is $k(x^2 - Sx + P)$, $k \neq 0$.
- Useful symmetric-function identities: $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$; $\alpha^3+\beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$; $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$.

Exercise 2.1

1. Find the zeros of each quadratic polynomial and verify the relationship between the zeros and coefficients: (i) $f(x) = x^2 - 2x - 8$ *[NCERT]* (ii) $g(s) = 4s^2 - 4s + 1$ *[NCERT]* (iii) $h(t) = t^2 - 15$ *[NCERT; CBSE 2024]*

2. If α, β are the zeros of $f(x) = x^2 - 5x + 4$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$.

3. If α, β are the zeros of $p(y) = y^2 - 7y + 1$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

4. If one zero of $f(x) = 4x^2 - 8kx - 9$ is the negative of the other, find k.

5. If the sum of the zeros of $f(t) = kt^2 + 2t + 3k$ equals their product, find k.

6. Find the zeros of each and verify the relationship with coefficients: (i) $p(x) = x^2 + 2\sqrt{2}x - 6$ (ii) $h(s) = 2s^2 - (1 + 2\sqrt{2})s + \sqrt{2}$ *[NCERT Exemplar]* (iii) $f(v) = v^2 + 4\sqrt{3}v - 15$ *[NCERT Exemplar]*

7. Find a quadratic polynomial whose sum and product of zeros, respectively, are: (i) $-2\sqrt{3}, -9$ (ii) $\frac{3}{2\sqrt{5}}, -\frac{1}{2}$ *[NCERT Exemplar]*

8. If α, β are the zeros of $p(x) = 4x^2 - 5x - 1$, find the value of $\alpha^2\beta + \alpha\beta^2$.

9. If α, β are the zeros of $f(x) = 6x^2 + x - 2$, find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

10. If α, β are the zeros of $f(x) = x^2 - p(x+1) - c$, show that $(\alpha+1)(\beta+1) = 1 - c$. *[HOTS]*

11. If α, β are zeros of a quadratic polynomial with $\alpha + \beta = 24$ and $\alpha - \beta = 8$, find the polynomial.

12. Find a quadratic polynomial whose zeros are the negatives of the zeros of $px^2 + qx + r$. *[HOTS]*

13. If α, β are the zeros of $f(x) = x^2 - 1$, find a quadratic polynomial whose zeros are $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$. *[HOTS]*

14. If the squared difference of the zeros of $f(x) = x^2 + px + 45$ is 144, find p. *[HOTS]*

15. If α, β are zeros of $f(x) = x^2 - px + q$, prove that $\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} = \frac{p^4}{q^2} - \frac{4p^2}{q} + 2$. *[HOTS]*

16. If α, β are the zeros of $f(x) = x^2 + x - 2$, find (i) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ *[CBSE 2024]* (ii) $\frac{1}{\alpha} - \frac{1}{\beta}$.

Answers: 1. (i) $4, -2$ (ii) $\frac{1}{2}, \frac{1}{2}$ (iii) $\sqrt{15}, -\sqrt{15}$. 2. $\frac{27}{4}$. 3. 7. 4. 0. 5. $\frac{2}{3}$. 6. (i) $-3\sqrt{2}, \sqrt{2}$ (ii) $\sqrt{2}, \frac{1}{\sqrt{2}}$ (iii) $\sqrt{3}, -5\sqrt{3}$. 7. (i) $f(x) = k(x^2 + 2\sqrt{3}x - 9)$; zeros $-3\sqrt{3}, \sqrt{3}$ (ii) $f(x) = k\left(x^2 - \frac{3}{2\sqrt{5}}x - \frac{1}{2}\right)$. 8. $-\frac{5}{16}$. 9. $-\frac{1}{6}$. 10. Proved. 11. $f(x) = k(x^2 - 24x + 128)$. 12. $px^2 - qx + r$. 13. $f(x) = k(x^2 + 4x + 4)$. 14. $p = \pm 18$. 15. Proved. 16. (i) $-\frac{5}{2}$ (ii) $\pm\frac{\sqrt{3}}{2}$.

Very Short Answer Questions (VSAQs)

1. The sum and product of the zeros of a quadratic polynomial are $\frac{1}{4}$ and -3 respectively. Find the polynomial.

2. If the product of the zeros of $f(x) = x^2 - 4x + k$ is 3, find k.

3. If the sum of the zeros of $f(x) = kx^2 - 3x + 5$ is 1, find k.

4. If $x = 1$ is a zero of $f(x) = x^3 - 2x^2 + 4x + k$, find k.

5. Write the zeros of $x^2 - x - 6$. *[CBSE 2008]*

6. If α, β are zeros of a polynomial with $\alpha + \beta = -b$ and $\alpha\beta = -4$, write the polynomial. *[CBSE 2010]*

7. For what value of k is 3 a zero of $2x^2 + x + k$? *[CBSE 2010]*

8. For what value of k is -3 a zero of $x^2 + 11x + k$? *[CBSE 2010]*

9. For what value of k is -2 a zero of $3x^2 + 4x + 2k$? *[CBSE 2010]*

10. If α, β are zeros of $p(x) = 5x^2 - 6x + 1$, find $\alpha + \beta + \alpha\beta$. *[CBSE 2024]*

11. If $(x + a)$ is a factor of $2x^2 + 2ax + 5x + 10$, find a. *[CBSE 2008]*

12. For what value of k is -4 a zero of $x^2 - x - (2k+2)$? *[CBSE 2009]*

13. If 1 is a zero of $p(x) = ax^2 - 3(a-1)x - 1$, find a.

14. If α, β are zeros of $2y^2 + 7y + 5$, write $\alpha + \beta + \alpha\beta$. *[CBSE 2010]*

15. If a quadratic polynomial factorises into distinct linear factors, how many real, distinct zeros does it have?

16. True or False: if a quadratic polynomial is the square of a linear polynomial, its two zeros are coincident.

17. If the graph of $ax^2 + bx + c$ cuts the negative y-axis, what is the sign of c? *[HOTS]*

18. If $f(a)f(b) < 0$ for a polynomial $f(x)$, how many zeros lie between a and b? *[HOTS]*

Answers: 1. $k\left(x^2 - \frac{1}{4}x - 3\right)$. 2. $k = 3$. 3. $k = 3$. 4. $k = -3$. 5. $3, -2$. 6. $f(x) = x^2 + bx - 4$. 7. -21. 8. 24. 9. -2. 10. $\frac{7}{5}$. 11. 2. 12. 9. 13. 1. 14. -1. 15. Two. 16. True. 17. Negative. 18. At least one.

Fill in the Blanks

1. Quadratic polynomials whose zeros are -4 and 3 are given by ________.

2. The zeros of $x^2 + x - 6$ are ________.

3. If $x + 1$ is a factor of $2x^3 + ax^2 + 4x + 1$, then $a =$________.

4. The graph of a quadratic polynomial intersects the x-axis at most at ________ points.

5. If -4 is a zero of $x^2 - x - (2k+2)$, then $k =$________.

6. If $4x^2 - 6x - m$ is divisible by $x - 3$, then $m =$________.

7. Quadratic polynomials with rational coefficients having $\sqrt{3}$ as a zero are given by ________.

8. If $x + a$ is a factor of $2x^2 + 2ax + 4x + 12$, then $a =$________.

9. The maximum number of zeros a quadratic polynomial can have is ________.

10. If $a + b + c = 0$, then one zero of $ax^2 + bx + c$ is ________. *[HOTS]*

11. If the zeros of $ax^2 + bx + c$ are both negative, then a, b, c all have the ________ sign.

Answers: 1. $k(x^2 + x - 12)$. 2. $-3, 2$. 3. 5. 4. Two. 5. 9. 6. 18. 7. $k(x^2 - 3)$. 8. 3. 9. Two. 10. 1. 11. Same.

CHAPTER 3

Pair of Linear Equations in Two Variables

*Per the CBSE 2026–27 syllabus, the **Cross-Multiplication Method** and **equations reducible to a pair of linear equations** (equations non-linear in x, y needing a substitution like $u = \frac{1}{x}$) have been removed. This edition's word-problem exercises use cross-multiplication only as an internal solving technique in the book's own worked examples — since the questions themselves only ask you to solve the resulting linear system, they remain valid and can equally be solved by substitution/elimination. Exercise items requiring $1/x, 1/y$ substitution have been excluded below.*

Key Concepts

- For $a_1x + b_1y + c_1 = 0,\ a_2x + b_2y + c_2 = 0$: unique solution if $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$; no solution (parallel) if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$; infinitely many (coincident) if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.
- Solve algebraically by **substitution** or **elimination**; solve graphically by plotting both lines.

Exercise 3.2 — Graphical Method

1. Solve graphically: $3x + y + 4 = 0$, $3x - y + 2 = 0$. *[CBSE 2023, 2024]*

2. Show graphically that $2x + 3y = 6$, $4x + 6y = 12$ has infinitely many solutions. *[CBSE 2010]*

3. Show graphically that $3x - 5y = 20$, $6x - 10y = -40$ is inconsistent.

Answers: 1. $x = -1, y = -1$. 2. Coincident lines, infinitely many solutions. 3. Parallel lines, no solution.

Exercise 3.3 — Algebraic Solution

1. Find x, y in a rectangle with sides $x + 3y$, 13, y, and (opposite side relations as per figure). *[NCERT Exemplar]*

2. Solve: $11x + 15y + 23 = 0$, $7x - 2y - 20 = 0$.

3. Solve: $99x + 101y = 499$, $101x + 99y = 501$.

4. Solve: $21x + 47y = 110$, $47x + 21y = 162$. *[NCERT Exemplar]*

5. Find the solution of $\frac{x}{10}+\frac{y}{5}-1=0$ and $\frac{x}{8}+\frac{y}{6}=15$. Hence find λ if $y=\lambda x+5$. *[NCERT Exemplar; HOTS]*

6. Write an equation of a line through the solution-point of $x+y=2$ and $2x-y=1$. How many such lines exist? *[NCERT Exemplar; HOTS]*

7. Write a pair of linear equations with unique solution $x=-1, y=3$. How many such pairs exist? *[NCERT Exemplar; HOTS]*

8. If $x+1$ is a factor of $2x^3+ax^2+2bx+1$, find a, b given $2a-3b=4$. *[NCERT Exemplar; HOTS]*

Answers: 1. $x=1, y=4$. 2. $x=2, y=-3$. 3. $x=3, y=2$. 4. $x=3, y=1$. 5. $x=340, y=-165, \lambda=-\frac{1}{2}$. 6. $3x+2y=5$; infinitely many. 7. E.g. $12x+5y=3,\ 3x+7y=18$; infinitely many. 8. $a=5, b=2$.

Exercise 3.4 — Consistency Conditions

1. Determine whether the system has a unique solution, no solution, or infinitely many: (i) $x-3y=3,\ 3x-9y=2$ (ii) $2x+y=5,\ 4x+2y=10$.

2. Find k for which $kx+2y=5,\ 3x+y=1$ has a unique solution.

3. Find k for which $2x+ky=1,\ 3x-5y=7$ has (i) a unique solution, (ii) no solution. Is there a k giving infinitely many solutions?

4. For what k do $2x+3y=2,\ (k+2)x+(2k+1)y=2(k-1)$ have infinitely many solutions? *[CBSE 2000, 2003]*

5. For what k does $2x+ky=11, 5x-7y=5$ have no solution?

6. Find c if $cx+3y+3-c=0, 12x+cy-c=0$ has infinitely many solutions. *[CBSE 2019]*

7. For which λ do $\lambda x+y=\lambda^2, x+\lambda y=1$ have (i) no solution? (ii) infinitely many? (iii) a unique solution? *[NCERT Exemplar]*

8. Find a, b for which the following have infinitely many solutions: (i) $(2a-1)x-3y=5,\ 3x+(b-2)y=3$ *[CBSE 2002C]* (ii) $2x-3y=7,\ (a+b)x-(a+b-3)y=4a+b$ *[NCERT]*

Answers: 1. (i) No solution (ii) infinitely many. 2. $k\neq 6$. 3. (i) $k\neq -\frac{10}{3}$ (ii) $k=-\frac{10}{3}$; no value gives infinitely many. 4. $k=4$. 5. $k=-\frac{14}{5}$. 6. $c=6$. 7. (i) $\lambda=-1$ (ii) $\lambda=1$ (iii) all real $\lambda\neq\pm 1$. 8. (i) $a=3, b=\frac{17}{5}$ (ii) $a=5, b=1$.

Exercise 3.5 — Word Problems: Costs & Investments

1. 7 audio cassettes and 3 video cassettes cost Rs 1110, while 5 audio and 4 video cost Rs 1350. Find the cost of each.

2. The coach of a cricket team buys 7 bats and 6 balls for Rs 3800; later 3 bats and 5 balls for Rs 1750. Find the cost of each. *[NCERT]*

3. A lending library has a fixed charge for the first three days and an additional charge per extra day. *[NCERT]*

4. Jamila sold a table and a chair for Rs 1050 at 10% and 25% profit respectively; at 25% and 10% profit respectively she'd have got Rs 1065. Find each cost price. *[NCERT Exemplar]*

5. Susan invested money in two schemes A and B at different interest rates. *[NCERT Exemplar]*

6. The cost of 4 pens and 4 pencil boxes is Rs 100; three times a pen's cost is Rs 15 more than a pencil box. Find each cost. *[NCERT Exemplar]*

7. Vijay divided bananas into two lots and sold them at different rates, with two different total-revenue scenarios. Find the total number of bananas. *[NCERT Exemplar]*

8. One friend says to the other, "Give me a hundred, and I'll be twice as rich as you." The other replies, "Give me ten, and I'll be six times as rich as you." Find their respective capitals. *[NCERT; HOTS]*

9. A says to B: "If you give me 30 of your mangoes, I will have twice as many left with you." B replies: "If you give me 10, I will have thrice as many left with you." How many mangoes does each have? *[HOTS]*

10. It takes 12 hours to fill a swimming pool using two pipes together. If the larger pipe runs for 4 hours and the smaller for 9 hours, only half the pool is filled. Find the time each pipe alone takes to fill the pool. *[NCERT Exemplar; HOTS]*

Answers: 1. Rs 30, Rs 300. 2. Bat Rs 500, ball Rs 50. 3. Fixed Rs 15, extra Rs 3/day. 4. Table Rs 500, chair Rs 400. 5. Rs 12000 in A, Rs 10000 in B. 6. Pen Rs 10, box Rs 15. 7. 500 bananas. 8. Rs 40, Rs 170. 9. $A : 34$, $B : 62$ (mangoes). 10. Larger pipe 20 hours, smaller pipe 30 hours.

Exercise 3.6 — Word Problems: Numbers & Digits

1. Half the difference of two numbers is 2; the greater plus twice the smaller is 13. Find the numbers. *[CBSE 2023]*

2. The sum of the digits of a two-digit number and the reversed number differ by 66, digits differ by 2; find the number(s). *[NCERT]*

3. A two-digit number is 4 times the sum of its digits and twice the product of its digits. Find it. *[CBSE 2005]*

4. A two-digit number: product of digits is 20; adding 9 reverses the digits. Find it. *[CBSE 2005]*

5. The sum of the digits of a two-digit number is 9; nine times the number is twice the reversed number. Find it. *[NCERT]*

6. Two numbers are in ratio 5 : 6; subtracting 8 from each changes the ratio to 4 : 5. Find the numbers. *[NCERT Exemplar]*

Answers: 1. $7, 3$. 2. 42 or 24. 3. 36. 4. 45. 5. 18. 6. $40, 48$.

Exercise 3.7 — Word Problems: Fractions

1. A fraction becomes $\frac{9}{11}$ if 2 is added to both numerator and denominator; becomes $\frac{5}{6}$ if 3 is added to both. Find it. *[NCERT]*

2. The sum of numerator and denominator of a fraction is 12; increasing the denominator by 3 makes it $\frac{1}{2}$. Find it. *[CBSE 2006C]*

Answers: 1. $\frac{7}{9}$. 2. $\frac{5}{7}$.

Exercise 3.8 — Word Problems: Ages

1. Five years ago, Nuri was thrice as old as Sonu; ten years later Nuri will be twice as old as Sonu. Find their ages. *[NCERT]*

2. Father's age is 3 times the sum of his two children's ages; after 5 years his age will be twice that sum. Find the father's age. *[CBSE 2003, 2019]*

3. Two years ago, a father was 5 times as old as his son; two years later his age will be 8 more than 3 times the son's age. Find present ages. *[CBSE 2004]*

4. Two years ago, Salim was thrice as old as his daughter; six years later he'll be four years older than twice her age. How old are they now? *[NCERT Exemplar]*

5. The father's age is twice the sum of his two children's ages; after 20 years his age equals their combined ages. Find the father's age. *[NCERT Exemplar]*

6. Three years ago, Rashmi was thrice as old as Nazma; ten years later Rashmi will be twice as old. Find their present ages. *[CBSE 2024]*

7. 16 years ago Ajay was 5 years older than his wife; their present ages are in ratio 8 : 9. Find their ages at marriage. *[CBSE 2024]*

Answers: 1. Nuri 50, Sonu 20. 2. Father's age = 45 years. 3. Father 42, son 10. 4. Salim 38, daughter 14. 5. Father's age = 40 years. 6. Rashmi 42, Nazma 16. 7. Ajay 35, wife 40 (present).

Exercise 3.9 — Word Problems: Speed, Distance, Time

1. Two people 16 km apart walk towards each other, meeting in 2 hours; walking the same direction, the faster catches up in 8 hours. Find their speeds. *[CBSE 2023]*

2. A boat goes 30 km upstream and 44 km downstream in 10 hours; 40 km upstream and 55 km downstream in 13 hours. Find the boat's speed and the stream's speed. *[CBSE 2019]*

3. Rowing at 5 km/h in still water takes thrice as long going 40 km upstream as downstream. Find the stream's speed. *[NCERT Exemplar]*

4. Ritu rows 20 km downstream in 2 hours and 4 km upstream in 2 hours. Find her speed in still water and the current's speed. *[NCERT]*

5. A motor boat travels 30 km upstream and 28 km downstream in 7 hours; 21 km upstream and return in 5 hours. Find the boat's and stream's speeds. *[NCERT Exemplar]*

6. A train travels a distance at uniform speed; had the speed been different, the time would differ by a stated amount. Find the distance. *[NCERT; CBSE 2020]*

Answers: 1. 5 km/h, 3 km/h. 2. Stream 3 km/h, boat 8 km/h. 3. 2.5 km/h. 4. Speed 6 km/h, current 4 km/h. 5. Boat 10 km/h, stream 4 km/h. 6. 600 km.

Exercise 3.10 — Miscellaneous Applications

1. ABCD is a cyclic quadrilateral with $\angle A = (4y + 20)^\circ$, $\angle B = (3y - 5)^\circ$, $\angle C = (4x)^\circ$, $\angle D = (7x + 5)^\circ$. Find the four angles. *[NCERT]*

2. Yash scored 40 marks getting 3 marks per correct answer and losing 1 per wrong; with a different marking scheme he'd score 50. Find the number of questions. *[NCERT]*

3. A part of hostel charges is fixed, the remainder depends on days stayed; two students' bills give two equations. Find the fixed charge and daily rate. *[NCERT; CBSE 2000]*

4. Two supplementary angles differ by 18°. Find them. *[NCERT; CBSE 2019]*

5. Meena withdrew Rs 2000 in Rs 50 and Rs 100 notes, receiving 25 notes in total. Find how many of each. *[NCERT; CBSE 2024]*

6. A shopkeeper gives books on rent: a fixed charge for the first two days, an additional charge per extra day. *[NCERT Exemplar]*

7. There are two examination rooms A and B; moving 10 candidates from A to B equalises them, or moving 20 from B to A doubles A relative to B. Find the number in each room. *[NCERT Exemplar]*

8. A railway half ticket costs half the full fare, with equal reservation charges. Find the full fare and reservation charge. *[NCERT Exemplar]*

9. A shopkeeper sells a saree at 8% profit and a sweater at 10% discount for Rs 1008 total; at 10% profit and 8% discount for Rs 1028. Find the cost price of the saree and the list price of the sweater. *[NCERT Exemplar]*

10. In a competitive exam, 1 mark per correct answer, $\frac{1}{2}$ deducted per wrong; total 120 questions attempted, 90 marks scored. Find the number correct. *[NCERT Exemplar]*

Answers: 1. $\angle A = 120°, \angle B = 70°, \angle C = 60°, \angle D = 110°$. 2. 20 questions. 3. Fixed Rs 400, per-day Rs 30. 4. $99°, 81°$. 5. 10 of Rs 50, 15 of Rs 100. 6. Rs 10 fixed, Rs 3/extra day. 7. Room A: 100, Room B: 80. 8. Fare Rs 210, reservation Rs 6. 9. Saree Rs 600, sweater Rs 400. 10. 100 correct.

Very Short Answer Questions (VSAQs)

1. Find k for which $x + y - 4 = 0$, $2x + ky - 3 = 0$ has no solution. *[CBSE 2020]*

2. Find k for which $2x - y = 5$, $6x + ky = 15$ has infinitely many solutions.

3. Find the number of solutions of $x + 2y - 8 = 0$, $2x + 4y = 16$. *[CBSE 2009]*

4. If $2x + y = 13$ and $4x - y = 17$, find $x - y$. *[CBSE 2024]*

5. The sum of two numbers is 105, their difference is 45. Find the numbers. *[CBSE 2024]*

6. Solve for x, y: $23x + 24y = 23$, $24x + 23y = 24$. *[CBSE 2024]*

Answers: 1. $k = 2$. 2. $k = -3$. 3. Infinitely many. 4. 2. 5. $75, 30$. 6. $x = 1, y = 0$.

Fill in the Blanks

1. The pair $y = 0$ and $y = -7$ has ________ solution.

2. If $3x + 2ky = 2$, $2x + 5y + 1 = 0$ has no solution, then $k =$________.

3. If a pair of linear equations is consistent, the lines are either ________ or ________.

4. If $\lambda x + 3y = 7$, $2x + 6y = 14$ has infinitely many solutions, then $\lambda =$________.

5. If $x = a, y = b$ solves $x - y = 2$, $x + y = 4$, then $a =$________, $b =$________.

Answers: 1. No. 2. $\frac{15}{4}$. 3. Intersecting; coincident. 4. $\lambda = 1$. 5. $a = 3,\ b = 1$.

CHAPTER 4

Quadratic Equations

Per the CBSE 2026–27 syllabus, ***equations reducible to quadratic form*** *(including rational equations with the variable in a denominator) and the method of* ***completing the square*** *have been removed. Exercise items requiring $1/x$-type manipulation have been excluded throughout; only equations already in (or directly expandable to) standard form $ax^2 + bx + c = 0$ are retained.*

Key Concepts

- Standard form: $ax^2 + bx + c = 0,\ a \neq 0$. Solve by **factorisation** or the **quadratic formula** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- **Discriminant** $D = b^2 - 4ac$: two distinct real roots if $D > 0$; equal roots if $D = 0$; no real roots if $D < 0$.

Exercise 4.1 — Identifying & Forming Quadratic Equations

1. Which of the following are quadratic equations? (i) $16x^2-3 = (2x+5)(5x-3)$ (ii) $\sqrt{3}x^2-2x+\dfrac{1}{3} = 0$

2. The product of two consecutive positive integers is 306. Form the quadratic equation to find the integers. *[NCERT]*

3. John and Jivanti together have 45 marbles; after each loses 5, the product of their remaining marbles is 124. Form the quadratic equation. *[NCERT]*

Exercise 4.3, 4.4 — Solving by Factorisation; Discriminant

1. Solve by factorisation: $x^2 + 11x + 3 = 0$. *[CBSE 2010]*

2. Solve by factorisation: $2x^2 + ax - a^2 = 0$. *[CBSE 2014]*

3. Solve by factorisation: $x^2 - 3abx + 2b^2 = 0$.

4. Solve by factorisation: $9x^2 - 6b^2x - (a^4 - b^4) = 0$. *[CBSE 2015]*

5. Solve by factorisation: $abx^2 + (b^2 - ac)x - bc = 0$. *[CBSE 2005]*

6. Solve by factorisation: $3\sqrt{5}x^2 + 25x - 10\sqrt{5} = 0$. *[CBSE 2015]*

7. Solve by factorisation: $3x^2 - 2\sqrt{6}x + 2 = 0$. *[NCERT; CBSE 2010, 2012]*

8. Write the discriminant of: (i) $(x-1)(2x-1)=0$ (ii) $x^2-2x+k=0$ (iii) $3x^2+2\sqrt{2}x-2\sqrt{3}=0$ (iv) $(x+5)^2=2(5x-3)$. *[CBSE 2019]*

9. Determine whether the following have real roots, and if so find them: (i) $x^2+4\sqrt{3}x-5=0$ (ii) $2x^2+5\sqrt{3}x+6=0$ (iii) $\sqrt{2}x^2+7x+5\sqrt{2}=0$ *[CBSE 2013; NCERT]* (iv) $2x^2-2\sqrt{2}x+1=0$ *[NCERT]*

Answers: 2. $-15, 7$. 3. $-2b, -b$. 4. $\frac{-b^2}{3}, \frac{2b^2}{3}$ (in terms of given parameters; see book for exact factorisation). 7. (i) 1 (ii) $4-4k$ (iii) 32 (iv) -124. 8. (i) $4\sqrt{3}, -5\sqrt{3}$ (ii) $-2\sqrt{3}, -\frac{\sqrt{3}}{2}$ (iii) $-\frac{\sqrt{2}}{2}, -4\sqrt{2}$ (iv) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$.

Exercise 4.5 — Nature of Roots

1. Determine the nature of the roots: (i) $2x^2-3x+5=0$ *[NCERT]* (ii) $2x^2-6x+3=0$ *[NCERT]* (iii) $3x^2-4\sqrt{3}x+4=0$ *[NCERT]* (iv) $4x^2+4\sqrt{3}x+3=0$ *[CBSE 2019]*

2. Find k for which the roots are real and equal: (i) $4x^2-2(k+1)x+(k+4)=0$ (ii) $4x^2-2(k+1)x+(k+1)=0$ *[CBSE 2017]*

3. Find k for equal roots: (i) $2x^2+kx+3=0$ *[NCERT]* (ii) $kx(x-2)+6=0$ *[NCERT; 2013]* (iii) $x^2-4kx+k=0$ *[CBSE 2012]*

4. Find k for which $(4-k)x^2+(2k+4)x+(8k+1)=0$ is a perfect square.

5. (i) Find k for which $(3k+1)x^2+2(k+1)x+1=0$ has equal roots; find the roots. *[CBSE 2014]* (ii) Find all k for which $x^2+kx+16=0$ has equal roots; find the roots. *[CBSE 2019]* (iii) Find p for which $px(x-2)+6=0$ has equal roots; find the roots. *[CBSE 2023]*

6. Find p for which $(2p+1)x^2-(7p+2)x+(7p-3)=0$ has equal roots; find these roots. *[CBSE 2014]*

7. Find p for which $(p+1)x^2-6(p+1)x+3(p+9)=0,\ p\neq-1$ has equal roots; find the roots. *[CBSE 2013, 2024]*

8. Find the least positive k for which $x^2+kx+4=0$ has real roots.

9. If the roots of $(b-c)x^2+(c-a)x+(a-b)=0$ are equal, prove that $2b=a+c$. *[CBSE 2002C]*

10. If the roots of $(1+m^2)x^2+2mcx+(c^2-a^2)=0$ are equal, prove that $c^2=a^2(1+m^2)$. *[CBSE 2017]*

11. Determine the nature of the roots: (i) $(x-2a)(x-2b)=4ab$ *[HOTS]* (ii) $2(a^2+b^2)x^2+2(a+b)x+1=0$ *[HOTS]*

12. Show that $2(a^2+b^2)x^2+2(a+b)x+1=0$ has no real roots when $a\neq b$. *[HOTS]*

13. If the roots of $ax^2+2bx+c=0$ and $bx^2-2\sqrt{ac}\,x+b=0$ are simultaneously real, prove $b^2=ac$. *[HOTS]*

14. Show that the roots of $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$ are always real, and equal only when $a=b=c$. *[HOTS]*

Answers: 1. (i) No real roots (ii) real, distinct (iii) equal roots (iv) equal roots. 2. (i) $k=3$ or $k=-5$ (ii) $k=0$ or $k=3$. 3. (i) $k=\pm 2\sqrt{6}$ (ii) $k=0$ or 6 (iii) $k=0$ or $\frac{1}{4}$. 4. $k=0$ or $k=3$. 5. (i) $k=\frac{1}{5}$, roots $-\frac{5}{4}$ each; or $k=1$ (ii) $k=\pm 8$ (iii) $p=\frac{2}{3}$. 6. $p=1$ or $p=\frac{3}{17}$. 7. $p=-1$ (excluded) or $p=2$. 8. $k=4$.

Exercise 4.6 — Word Problems: Numbers

1. The sum of the squares of two consecutive odd positive integers is 394. Find them. *[CBSE 2009, 2017]*

2. Two numbers differ by 3 and their product is 504. Find them. *[CBSE 2002C]*

3. Three consecutive positive integers: sum of the square of the first and product of the other two is 46. Find them. *[CBSE 2010]*

4. The difference of squares of two numbers is 88; the larger is 5 less than twice the smaller. Find them. *[CBSE 2010]*

5. The sum of the squares of two consecutive multiples of 7 is 637. Find them. *[CBSE 2014]*

6. The sum of the squares of two consecutive even numbers is 340. Find them. *[CBSE 2014]*

7. The difference of the squares of two positive integers is 180; the square of the smaller is 8 times the larger. Find them. *[NCERT; CBSE 2014, 2022]*

8. Find a natural number whose square diminished by 84 equals thrice of 8 more than the number. *[NCERT Exemplar]*

Answers: 1. $13, 15$. 2. $21, 24$ (or $-24, -21$). 3. $5, 6, 7$. 4. $9, 13$. 5. $14, 21$. 6. $12, 14$. 7. $12, 18$. 8. 12.

Exercise 4.7 — Word Problems: Speed, Distance, Time

1. A train covers 90 km at uniform speed; 15 km/h more would take 30 min less. Find the original speed. *[CBSE 2006C, 2024]*

2. A train travels 360 km at uniform speed; 5 km/h more would take 1 hour less. Find the speed. *[NCERT]*

3. A plane's average speed problem: find the speed given a time-difference condition. *[NCERT Exemplar]*

4. Find a train's original average speed given a delay and a make-up-speed condition for the rest of the journey. *[NCERT Exemplar; CBSE 2018]*

5. Find the speed of a stream, given a boat's upstream/downstream time relationship. *[CBSE 2014, 2018]*

Exercise 4.8 — Word Problems: Ages

1. Ashu is x years old, his mother x^2 years old; in 5 years the mother will be 3 times Ashu's age. Find their present ages.

2. The sum of the ages of a man and his son is 45 years; 5 years ago, the product of their ages was 4 times the man's age then. Find their present ages.

3. Two years ago, a man's age was 3 times the square of his son's age; find their present ages. *[NCERT Exemplar]*

4. At present, Asha's age (in years) is 2 more than the square of her daughter Nisha's age; find both present ages. *[NCERT Exemplar]*

5. A man's age problem leading to a quadratic; find their present ages. *[CBSE 2024]*

Exercise 4.9 — Word Problems: Geometry (Pythagoras)

1. The hypotenuse of a right triangle is 25 cm; the other two sides differ by 5 cm. Find them.

2. The diagonal of a rectangular field is 60 m more than the shorter side; the longer side is 30 m more than the shorter. Find the sides.

3. Find the base of a triangle given its area and a relation between base and height. *[CBSE 2015]*

Exercise 4.10 — Word Problems: Area & Perimeter

1. Is it possible to design a rectangular mango grove of length twice its breadth and area 800 m^2? If so, find the dimensions. *[NCERT]*

2. Is it possible to design a rectangular park of perimeter 80 m and area 400 m^2? If so, find the dimensions.

3. Find the sides of two squares given a relation between their areas and side-lengths. *[NCERT; CBSE 2008, 2013]*

4. In the centre of a rectangular lawn, a rectangular pond is to be built; find its dimensions given the surrounding grass area. *[NCERT Exemplar]*

Exercise 4.11 — Word Problems: Time & Work

1. A takes 10 days less than B to finish a work; together they finish it in 12 days. Find the time each takes.

2. Two pipes together fill a reservoir in 12 hours; one alone takes 10 hours less than the other. Find each pipe's time. *[CBSE 2019, 2023]*

3. Two water taps together fill a tank in a given time; find the time each tap takes separately. *[NCERT]*

Exercise 4.12 — Miscellaneous Word Problems

1. A piece of cloth costs Rs 35; if it were 4 m longer and each metre cost Re 1 less, the total cost would stay the same. Find its length.

2. A picnic's food budget was Rs 480; 8 students dropped out, raising each remaining student's share by Rs 10. Find the original number of students.

3. Reducing a toy's list price by Rs 2 lets a buyer purchase 2 more toys for Rs 360. Find the original price.

4. Rashi got a total of 30 marks in Maths and Science; had she got 2 more in Maths and 3 less in Science, the product would have been 210. Find her marks in each subject. *[NCERT; CBSE 2008, 2014, 2019]*

5. At t minutes past 2 p.m., the time needed by the minute hand to show 3 p.m. was 3 minutes less than $\frac{t^2}{4}$ minutes. Find t. *[NCERT Exemplar]*

Very Short Answer Questions (VSAQs)

1. Write the standard form of a quadratic equation.

2. If the sum of the roots of $ax^2+bx+c=0$ is equal to the sum of the squares of their reciprocals, write the relation among a, b, c. *[HOTS]*

3. If the sum and product of the roots of $kx^2+6x+4k=0$ are equal, find k.

4. If -5 is a root of $2x^2+px-15=0$, find p.

5. If one root of $5x^2+13x+k=0$ is the reciprocal of the other, find k.

6. If $1+\sqrt{2}$ is a root of a quadratic equation with rational coefficients, write its other root. *[HOTS]*

7. Write the number of real roots of $x^2+3|x|+2=0$. *[HOTS]*

8. Write the value of λ for which $x^2+4x+\lambda$ is a perfect square. *[HOTS]*

9. If $mx^2+2x+m=0$ has only one real value of x satisfying it, find the values of m. *[HOTS]*

Answers: 1. Write the standard form: $ax^2+bx+c=0,\ a\neq 0$. 2. $2a^2c=ab^2+bc^2$. 3. $k=-\frac{3}{2}$. 4. $p=7$. 5. $k=5$. 6. Other root $=1-\sqrt{2}$. 7. No real roots (0). 8. $\lambda=4$. 9. $m=1$ or -1.

CHAPTER 5

Arithmetic Progressions

No topics have been removed from Arithmetic Progressions in the CBSE 2026–27 syllabus; this chapter is retained in full.

Key Concepts

- n^{th} term: $a_n = a + (n-1)d$. Sum of first n terms: $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$.
- $a_n = S_n - S_{n-1}$.

Exercise 5.2, 5.3 — Identifying an AP; First Term & Common Difference

1. The n^{th} term of an AP is $6n + 2$. Find the common difference. *[CBSE 2008]*

2. Show that $a_n = 5n - 7$ defines an AP; find its common difference.

3. The n^{th} term of an AP cannot be $n^2 + 1$. Justify your answer. *[NCERT Exemplar]*

4. Write the first term and common difference of: (i) $-5, -1, 3, 7, \ldots$ *[NCERT]* (ii) $\frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{7}{5}, \ldots$

Answers: 1. $d = 6$. 2. $d = 5$. 4. (i) $a = -5, d = 4$ (ii) $a = \frac{2}{5}, d = \frac{1}{5}$.

Exercise 5.4 — The n^{th} Term

1. Find: (i) 10^{th} term of $1, 4, 7, 10, \ldots$ (ii) 18^{th} term of $\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}, \ldots$ (iii) n^{th} term of $13, 8, 3, -2, \ldots$

2. (i) Which term of $3, 8, 13, \ldots$ is 248? (ii) Which term of $65, 61, 57, \ldots$ is its first negative term? *[CBSE 2023]*

3. (i) Is 302 a term of $3, 8, 13, \ldots$? (ii) Is -150 a term of $11, 8, 5, 2, \ldots$? *[CBSE 2017]*

4. Find the number of terms in: (i) $3, 8, 13, \ldots, 253$ *[NCERT]* (ii) $1, 4, 7, 10, \ldots, 88$

5. Find whether 0 is a term of the AP $40, 37, 34, 31, \ldots$. *[CBSE 2014]*

6. If the 7^{th} term of an AP is $\frac{1}{9}$ and the 9^{th} term is $\frac{1}{7}$, find its 63^{rd} term. *[CBSE 2014]*

7. Which term of $3, 15, 27, 39, \dots$ will be 120 more than its 21^{st} term? *[CBSE 2009]*

8. How many multiples of 4 lie between 10 and 250? *[NCERT]*

9. How many three-digit numbers are divisible by 7? *[CBSE 2013; NCERT]*

10. Find the number of all three-digit natural numbers divisible by 9. *[CBSE 2013]*

11. How many numbers lie between 10 and 300 which, when divided by 4, leave remainder 3? *[NCERT Exemplar; HOTS]*

12. If the ratio of the 11^{th} term to the 18^{th} term of an AP is $2 : 3$, find the ratio of the 5^{th} to the 21^{st} term.

13. If m times the m^{th} term of an AP equals n times its n^{th} term, show that the $(m+n)^{\text{th}}$ term is zero. *[CBSE 2008, 2019]*

14. If the $(m+1)^{\text{th}}$ term of an AP is twice the $(n+1)^{\text{th}}$ term, prove that the $(3m+1)^{\text{th}}$ term is twice the $(m+n+1)^{\text{th}}$ term. *[HOTS]*

15. If an AP has n terms with first term a and last term l, show that the sum of the m^{th} term from the beginning and the m^{th} term from the end is $(a+l)$. *[HOTS]*

Answers: 1. (i) 28 (ii) $35\sqrt{2}$ (iii) $18 - 5n$. 2. (i) 50^{th} (ii) 17^{th} term, value -3. 3. (i) Yes, 60^{th} term (ii) No. 4. (i) 51 (ii) 30. 5. No. 6. 1. 7. 31^{st} term. 8. 60. 9. 128. 10. 100. 11. 73 numbers.

Exercise 5.5 — Numbers/Terms in AP; Word Problems

1. Find x for which $(8x+4), (6x-2), (2x+7)$ are in AP.

2. Show that $(a-b)^2, (a^2+b^2), (a+b)^2$ are in AP. *[CBSE 2020]*

3. Three numbers in AP have sum 27 and product 648. Find them.

4. The sum of three terms of an AP is 21, and the product of the first and third exceeds the second by a stated amount; find the numbers. *[CBSE 2024]*

5. Split 207 into three parts in AP such that the product of the two smaller parts is 4623. *[NCERT Exemplar; HOTS]*

6. The angles of a quadrilateral are in AP with common difference 10°. Find the angles. *[HOTS]*

7. The angles of a triangle are in AP; the greatest is twice the least. Find all three angles. *[NCERT Exemplar; HOTS]*

8. In an AP, the sum of the first n terms is $\dfrac{3n^2}{2} + \dfrac{5n}{2}$. Find its 25^{th} term. *[CBSE 2006C]*

9. How many terms of an AP must be taken for their sum to reach a stated value, explaining the double answer where applicable. *[CBSE 2005]*

10. A man received a scholarship increasing by a fixed amount each year; find the amounts in specific years. *[CBSE 2015; NCERT]*

11. Sub-divisional officer plants trees such that each section of each class plants as many trees as the class number, for 1 to 12; find the total trees planted. *[CBSE 2014; NCERT]*

Answers: 1. $x = \frac{15}{4}$. 3. $4, 9, 14$ (or $14, 9, 4$). 5. $67, 69, 71$. 6. $75°, 85°, 95°, 105°$. 7. $40°, 60°, 80°$.

Exercise 5.6 — Sum of n Terms

1. Find the sum: (i) $-26, -24, -22, \dots$ to 36 terms (ii) $a + b, a - b, a - 3b, \dots$ to 22 terms

2. Find the sum of the last ten terms of the AP $8, 10, 12, \dots, 126$. *[NCERT Exemplar]*

3. Find S_n if $a_n = 3 + 4n$; also if $a_n = 9 - 5n$. *[NCERT]*

4. How many terms of the AP $9, 17, 25, \dots$ must be taken so their sum is 636? *[NCERT]*

5. Find the sum of: (i) the first 15 multiples of 8 *[NCERT; CBSE 2017]* (ii) the first 40 positive integers divisible by 6 *[NCERT]* (iii) all 3-digit natural numbers divisible by 13 *[CBSE 2006C]* (iv) all 3-digit multiples of 11 *[CBSE 2012]* (v) all 2-digit natural numbers divisible by 4 *[CBSE 2017]*

6. Find the sum: (i) $34 + 32 + 30 + \cdots + 10$ (ii) $25 + 28 + 31 + \cdots + 100$ *[CBSE 2006C]*

7. Find the common difference of an AP whose first term is given along with a sum condition. *[CBSE 2019]*

8. Solve for x: $(-4) + (-1) + 2 + 5 + \cdots + x = 437$. *[NCERT Exemplar; HOTS]*

9. The sum of the first n terms of an AP (first term 8, common difference 20) equals the sum of the first $2n$ terms of another AP (first term -30, common difference 8). Find n. *[NCERT Exemplar; HOTS]*

10. 27 flags are fixed at 2 m intervals along a straight passage, stored at the middle flag's position. Ruchi carries one flag at a time. Find the total distance covered (including returning for her books), and the maximum distance carrying a single flag. *[NCERT Exemplar; HOTS]*

Answers: 2. 1170. 4. 12 terms. 8. $x = 50$. 9. $n = 11$. 10. Total distance $= 728$ m; maximum carrying distance $= 26$ m.

Very Short Answer Questions (VSAQs)

1. The 11$^{\text{th}}$ term of the AP $-5, -\frac{5}{2}, 0, \frac{5}{2}, \dots$ is ________. *[NCERT Exemplar]*

2. The 21$^{\text{st}}$ term of the AP whose first two terms are -3 and 4 is ________. *[NCERT Exemplar]*

3. Find the common difference of the AP for which a given sum-of-terms relation is stated. *[CBSE 2019]*

Answers: 1. 20. 2. 137.

CHAPTER 6

Co-ordinate Geometry

Per the CBSE 2026–27 syllabus, the coordinate ***Area of a Triangle*** *formula (the determinant/"shoelace" formula) has been removed. Questions below that ask for the area of a triangle, rhombus, or rectangle are retained only where the area can be found via the Distance Formula and elementary area formulas (e.g. area of a right triangle =* $\frac{1}{2}\times$ *base* $\times$ *height, or area of a rhombus =* $\frac{1}{2}d_1d_2$*) rather than the coordinate area formula.*

Key Concepts

- Distance formula: $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Section formula (ratio $m_1 : m_2$): $\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$; mid-point: $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$.
- Centroid of a triangle: $\left(\dfrac{x_1 + x_2 + x_3}{3}, \dfrac{y_1 + y_2 + y_3}{3}\right)$.

Exercise 6.2 — Distance Formula

1. Find the distance between: (i) $(-6, 7)$ and $(-1, -5)$ (ii) $(a + b, b + c)$ and $(a - b, c - b)$

2. Find a when the distance between $(3, a)$ and $(4, 1)$ is $\sqrt{10}$.

3. Show that $(-4, -1), (-2, -4), (4, 0), (2, 3)$ are the vertices of a rectangle. *[CBSE 2006C]*

4. Show $\triangle ABC$ with $A(-2, 0), B(2, 0), C(0, 2)$ and $\triangle PQR$ with $P(-4, 0), Q(4, 0), R(0, 4)$ are similar. *[CBSE 2017]*

5. Prove that $(3, 0), (6, 4), (-1, 3)$ are vertices of a right-angled isosceles triangle. *[CBSE 2006C, 2013]*

6. Prove $(2, -2), (-2, 1), (5, 2)$ are vertices of a right triangle. Find its area and hypotenuse. *[CBSE 2016]*

7. $A(2, 9), B(a, 5), C(5, 5)$ are vertices of $\triangle ABC$ right-angled at B. Find a and hence the area. *[NCERT Exemplar]*

8. If $P(2, 2)$ is equidistant from $A(-2, k)$ and $B(-2k, -3)$, find k and the length AP. *[CBSE 2014]*

9. Find a relation between x, y such that (x, y) is equidistant from $(3, 6)$ and $(-3, 4)$. *[NCERT; CBSE 2024]*

10. Prove that $(-2, 5), (0, 1), (2, -3)$ are collinear.

11. If $A(2, -4)$ is equidistant from $P(3, 8)$ and $Q(-10, y)$, find y and the distance PQ. *[NCERT Exemplar]*

12. Three vertices of a parallelogram are $(3, 4), (3, 8), (9, 8)$. Find the fourth.

13. Name the quadrilateral formed by $A(2, -2), B(7, 3), C(1, -1), D(6, -6)$, with reasons. *[NCERT Exemplar]*

14. Prove that $(3, 0), (4, 5), (-1, 4), (-2, -1)$, taken in order, form a rhombus; find its area. *[NCERT]*

15. Find a point on the y-axis equidistant from $(5, -2)$ and $(-3, 2)$. *[CBSE 2009, 2019]*

16. Find a point on the x-axis equidistant from $(7, 6)$ and $(-3, 4)$. *[CBSE 2005]*

17. Prove $A(2, 3), B(-2, 2), C(-1, -2), D(3, -1)$ are vertices of square $ABCD$. *[CBSE 2013]*

18. If $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$, find x; also find QR and PR. *[NCERT]*

19. Find y for which the distance between $P(2, -3)$ and $Q(10, y)$ is 10 units. *[NCERT]*

20. Find the equation of the perpendicular bisector of the segment joining $(7, 1)$ and $(3, 5)$. *[NCERT]*

21. The centre of a circle is $(2a, a - 7)$; find a if the circle passes through $(11, -9)$ with diameter $10\sqrt{2}$. *[NCERT Exemplar]*

22. Ayush walks house→bank→school→office instead of directly; find the extra distance. Coordinates: house $(2, 4)$, bank $(5, 8)$, school $(13, 14)$, office $(13, 26)$, in km. *[NCERT Exemplar]*

23. If $(0, -3)$ and $(0, 3)$ are two vertices of an equilateral triangle, find the third vertex. *[CBSE 2014]*

24. An equilateral triangle has two vertices at $(3, 4)$ and $(-2, 3)$; find the third vertex. *[HOTS]*

25. Find the circumcentre of the triangle whose vertices are $(-2, -3), (-1, 0), (7, -6)$. *[HOTS]*

26. If $P(-3, 4)$ is reflected in the x-axis to Q and in the y-axis to R, find the length QR. *[HOTS]*

Answers: 1. (i) 13 (ii) $2\sqrt{2}\,|b|$. 2. $a = 4$ or -2. 6. Area $= \dfrac{15}{2}$ sq. units, hypotenuse $= 5\sqrt{2}$. 7. $a = 2$, area $= \dfrac{17}{2}$ sq. units. 8. $k = -1$ or 3. 11. $y = -3$ or -5; $PQ = \sqrt{290}$ or $13\sqrt{2}$. 12. $(9, 4)$. 23. Third vertex $(3\sqrt{3}, 0)$ or $(-3\sqrt{3}, 0)$. 25. Circumcentre $= (3, -3)$. 26. $Q(-3, -4), R(3, 4)$, giving $QR = \sqrt{36 + 64} = 10$.

Exercise 6.3 — Section Formula

1. Find the point dividing $(-1, 3)$ and $(4, -7)$ internally in ratio $3 : 4$.

2. Find where the diagonals of the parallelogram formed by $(-2, -1), (1, 0), (4, 3), (1, 2)$ meet.

3. $A(3,1), B(5,1), C(a,b), D(4,3)$ are vertices of parallelogram $ABCD$. Find a, b. *[CBSE 2019]*

4. Find the ratio in which $(2, y)$ divides $A(-2,2), B(3,7)$; find y. *[CBSE 2009]*

5. If $A(-1,3), B(1,-1), C(5,1)$ are vertices of $\triangle ABC$, find the length of the median through A.

6. Show that the diagonals of quadrilateral $PQRS$ bisect each other (given vertices). *[CBSE 2024]*

7. Find the ratio in which $(-4,6)$ divides the segment joining $A(-6,10)$ and $B(3,-8)$. *[NCERT; CBSE 2015, 2017]*

8. Find the ratio in which the x-axis divides a given segment; find the point of division. *[CBSE 2006, 2010, 2016, 2019, 2023]*

9. In what ratio does the point R divide PQ such that $PR = 2QR$? *[CBSE 2024]*

10. Find y given a section-formula condition with points including $B(6,-3)$. *[CBSE 2004, 2018]*

11. If (a, b) is the mid-point of a segment $A(10,-6)$, $B(k, 4)$ and $a - 2b = 18$, find k and AB. *[NCERT Exemplar]*

12. Find the coordinates of P such that $AP = \frac{2}{5}AB$. *[NCERT; CBSE 2008, 2009]*

Answers: 1. $\left(\frac{5}{7}, -1\right)$. 4. Ratio $4:1$, $y = 6$.

Exercise 6.4 — Centroid

1. Find the centroid of the triangle with vertices: (i) $(1,4), (-1,-1), (3,-2)$ (ii) $(-2,3), (2,-1), (4,0)$.

2. Two vertices of a triangle are $(1,2), (3,5)$; centroid is at the origin. Find the third vertex. *[HOTS]*

3. Find the third vertex given two vertices $(-3,1), (0,-2)$ and centroid at the origin. *[HOTS]*

4. $A(3,2), B(-2,1)$ are two vertices of $\triangle ABC$ with centroid $\left(\frac{5}{3}, -\frac{1}{3}\right)$. Find C. *[CBSE 2004]*

5. If $(-2,3), (4,-3), (4,5)$ are the mid-points of the sides of a triangle, find its centroid.

6. $A(x_1,y_1), B(x_2,y_2), C(x_3,y_3)$ are vertices of $\triangle ABC$. (i) The median from A meets BC at D; find D. (ii) Find P on AD with $AP : PD = 2 : 1$. (iii) Find Q, R on medians BE, CF with $BQ : QE = 2 : 1$, $CR : RF = 2 : 1$. (iv) State the centroid's coordinates. *[NCERT Exemplar; HOTS]*

Answers: 1. (i) $\left(1, \frac{1}{3}\right)$ (ii) $\left(\frac{4}{3}, \frac{2}{3}\right)$. 2. $(-4,-7)$. 3. $(3,1)$. 4. $(4,-4)$. 5. $\left(2, \frac{5}{3}\right)$. 6. (i) $D = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$ (ii)–(iv) P, Q, R all coincide with the centroid $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$.

Very Short Answer Questions (VSAQs)

1. Find the distance of $P(x, y)$ from the origin. *[CBSE 2018]*

2. Find P on the x-axis equidistant from $A(-2, 0)$ and $B(6, 0)$. *[CBSE 2019]*

3. Find x if the distance between $(0, 0)$ and $(x, -4)$ is 5. *[CBSE 2019]*

4. If the distance between $(4, k)$ and $(1, 0)$ is 5, find possible k. *[CBSE 2017]*

5. $A(1, 2), B(4, 3), C(6, 6)$ are three vertices of parallelogram $ABCD$; find D. *[CBSE 2010]*

6. If $P(2, p)$ is the mid-point of $A(6, -5)$ and $B(-2, 11)$, find p. *[CBSE 2010]*

7. If the distance between $(3, 0)$ and $(0, y)$ is 5 and $y > 0$, find y. *[CBSE 2010]*

8. What is the distance between $A(c, 0)$ and $B(0, -c)$? *[CBSE 2010]*

9. Find a so that $(3, a)$ lies on $2x - 3y - 5 = 0$. *[CBSE 2009]*

10. Find the ratio in which the x-axis divides the segment joining $A(3, -6)$ and $B(5, 3)$.

11. P divides $A(4, -5), B(4, 5)$ such that $AP : AB = 2 : 5$. Find P. *[CBSE 2023]*

12. If $P(x, y)$ is equidistant from $A(5, 1)$ and $B(1, 5)$, prove $x = y$. *[CBSE 2023]*

Answers: 1. $\sqrt{x^2 + y^2}$. 2. $(2, 0)$. 3. $x = \pm 3$. 4. $k = 0$ or 8. 5. $D(3, 5)$. 6. $p = 3$. 7. $y = 4$. 8. $c\sqrt{2}$. 9. $a = \frac{1}{3}$. 10. Ratio $2 : 1$. 11. $P(4, -1)$.

CHAPTER 7

Triangles

Per the CBSE 2026–27 syllabus, formal proofs *of the Basic Proportionality Theorem (BPT) and the Pythagoras Theorem are excluded; all problems below* apply *these results to solve numerical or short-proof problems, which remains fully in syllabus. Several figure-based items are paraphrased in words below in place of the original diagram.*

Key Concepts

- **Similarity criteria:** AAA (AA), SSS, SAS.
- **BPT (Thales):** a line parallel to one side of a triangle divides the other two sides in the same ratio; and its converse.
- **Angle Bisector Theorem:** the internal bisector of an angle of a triangle divides the opposite side in the ratio of the sides containing the angle.
- Ratio of areas of similar triangles = (ratio of corresponding sides)^2.
- **Pythagoras Theorem** and its converse (used as tools; formal proof not examined).

Exercise 7.1 — Similarity Basics

1. Fill in the blanks: (i) all circles are _____ (ii) all squares are _____ (iii) two triangles are similar if their corresponding angles are _____ and sides are _____. *[NCERT]*

2. State True/False with reasons: (i) any two similar figures are congruent (ii) any two congruent figures are similar (iii) two polygons are similar if corresponding sides are proportional. *[NCERT Exemplar]*

Exercise 7.2 — Basic Proportionality Theorem: Applications

1. In $\triangle ABC$, D, E on AB, AC with $DE \parallel BC$: (i) $AD = 6, DB = 9, AE = 8$; find AC (ii) $\dfrac{AD}{DB} = \dfrac{3}{4}$, $AC = 15$; find AE (iii) $AD = 2.5, BD = 3.0, AE = 3.75$; find AC. *[CBSE 2006C]*

2. Find x for which $DE \parallel AB$, given expressions for AD, DB, AE, EC in terms of x. *[NCERT Exemplar]*

3. D, E on sides AB, AC of $\triangle ABC$ with $AD = BD$ and $AE = CE$... if $DE \parallel BC$ with given lengths, find the unknown. *[CBSE 2023]*

4. If D, E are points on AB, AC such that $BD = CE$, prove $\triangle ABC$ is isosceles (given $DE \parallel BC$). *[CBSE 2009]*

5. $ABCD$ is a trapezium with $AB \parallel DC$; diagonals meet at O. Using BPT, prove $\dfrac{OA}{OC} = \dfrac{OB}{OD}$. *[CBSE 2010, 2023]*

6. D is a point on side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \times CD$. *[NCERT; CBSE 2004, 2023]*

7. If a line intersects sides AB and AC of $\triangle ABC$ at D, E and is parallel to BC, prove $\dfrac{AD}{AB} = \dfrac{AE}{AC}$.

8. O is the point of intersection of diagonals of a trapezium $PQRS$ with $PQ \parallel RS$; a line through O parallel to PQ meets PS in M; prove MO bisects the third diagonal-related segment (as per figure). *[NCERT]*

Exercise 7.3 — Angle Bisector Theorem

1. In $\triangle ABC$, AD bisects $\angle A$, meeting BC at D: (i) $BD = 2.5, AB = 5, AC = 4.2$; find DC (ii) $BD = 2, AB = 5, DC = 3$; find AC. *[CBSE 2001]*

2. AE bisects the exterior $\angle CAD$ meeting BC produced at E; given AB, AC, BC, find the required length.

3. Check whether AD bisects $\angle A$ of $\triangle ABC$: (i) $AB = 4, AC = 6, BD = 1.6, CD = 2.4$ (ii) $AB = 8, AC = 24, BD = 6, BC = 24$.

Exercise 7.4 — Similarity Criteria & Pythagoras Applications

1. $\triangle ABC \sim \triangle DEF$ given various side/angle data; find the requested unknown sides. *[NCERT Exemplar]*

2. If $\triangle ABC \sim \triangle DEF$, $BC = 3$ cm, $EF = 4$ cm and area($\triangle ABC$) = 54 cm^2, find area($\triangle DEF$). *[NCERT Exemplar]*

3. Corresponding sides of two similar triangles are in ratio 4 : 9; find the ratio of their areas.

4. Diagonals of a trapezium $ABCD$ with $AB \parallel CD$ intersect at O; if $AB = 2CD$, find the ratio of areas of $\triangle AOB$ and $\triangle COD$. *[NCERT Exemplar]*

5. $\triangle ABC$ is right-angled at B; $BD \perp AC$. If $AD = 4$ cm, $CD = 5$ cm, find BD and AB. *[NCERT Exemplar]*

6. A street light bulb on a pole 6 m high; a woman 1.5 m tall casts a shadow 3 m long. Find her distance from the pole. *[NCERT]*

7. In quadrilateral $ABCD$ with diagonals AC, PQ intersecting at O, $AB \parallel DC$; prove $OA \cdot CQ = OC \cdot AP$. *[NCERT Exemplar]*

8. If $AD \perp BC$ and $AD^2 = BD \times DC$, prove $\angle BAC = 90°$. *[CBSE 2024]*

9. O is a point inside a rectangle $ABCD$; prove $OB^2 + OD^2 = OA^2 + OC^2$. *[NCERT Exemplar]*

10. A 15 m high tower casts a shadow 24 m long; find the height of a pole casting a 16 m shadow at the same time. *[NCERT Exemplar]*

11. $AB \perp BC$, $DE \perp AC$; prove $\triangle ABC \sim \triangle AED$. *[CBSE 2009]*

12. Prove $\triangle ABE \sim \triangle CFB$ (given right-angle and parallel conditions). *[NCERT; CBSE 2008, 2024]*

13. Find the ratio of the areas of two similar triangles given a ratio of their corresponding medians. *[NCERT; CBSE 2019]*

14. In an equilateral triangle, prove that three times the square of one side equals four times the square of one altitude.

15. In a right triangle, the square of the hypotenuse equals the sum of squares of the other two sides — verify with a given numerical triangle and classify it as right-angled or not. *[NCERT, CBSE 2023, 2024]*

16. $ABCD$ is a quadrilateral; P, Q, R, S are points of trisection of AB, BC, CD, DA respectively, adjacent to A and C. Prove $PQRS$ is a parallelogram. *[HOTS]*

17. In quadrilateral $ABCD$, if the bisectors of $\angle ABC$ and $\angle ADC$ meet on diagonal AC, prove the bisectors of $\angle BAD$ and $\angle BCD$ meet on diagonal BD. *[HOTS]*

18. If $\dfrac{OA}{OC} = \dfrac{OP}{OB}$ with vertically opposite angles at O, prove $\angle A = \angle C$ and $\angle B = \angle D$. *[NCERT; HOTS]*

19. $\triangle FEC \cong \triangle GBD$ and $\angle 1 = \angle 2$; prove $\triangle ADE \sim \triangle ABC$. *[CBSE 2024; HOTS]*

20. PA, QB, RC are each perpendicular to AC (with P, Q, R suitably placed); prove $\dfrac{1}{x} + \dfrac{1}{z} = \dfrac{1}{y}$ where x, y, z are the respective perpendicular lengths. *[CBSE 2024; HOTS]*

Answers: Ex.7.2(1): (i) $AC = 14$ cm (ii) $AE = \dfrac{45}{7}$ cm (iii) $AC = 6.25$ cm. Ex.7.4(2): area $= 96\ \text{cm}^2$. (3): ratio 16 : 81. (4): ratio 4 : 1. (5): $BD = 2\sqrt{5}$ cm, $AB = 6$ cm. (6): 9 m. (10): 10 m.

Very Short Answer Questions (VSAQs)

1. State the basic proportionality theorem (statement only, no proof required).

2. In $\triangle ABC$, $DE \parallel BC$, $AD = x, DB = x - 2, AE = x + 2, EC = x - 1$; find x. *[CBSE 2023]*

3. If $\triangle ABC \sim \triangle PQR$ with $\dfrac{AB}{PQ} = \dfrac{1}{3}$, find $\dfrac{\text{ar}(ABC)}{\text{ar}(PQR)}$.

4. The altitude of an equilateral triangle of side 8 cm is ________.

5. If $\triangle ABC \sim \triangle DEF$ such that $2AB = DE$ and $BC = 8$ cm, find EF.

Answers: 2. $x = 4$. 3. $\dfrac{1}{9}$. 4. $4\sqrt{3}$ cm. 5. 16 cm.

CHAPTER 8

Circles

No topics have been removed from Circles in the CBSE 2026–27 syllabus; the tangent-length and tangent-perpendicular-to-radius results are used throughout as tools. Several figure-based items are paraphrased in words below.

Key Concepts

- A tangent to a circle is perpendicular to the radius at the point of contact.
- From an external point, exactly two tangents can be drawn to a circle, and their lengths are equal.

Exercise 8.1

1. O is the centre of a circle of radius 8 cm; the tangent at A cuts a line through O at B with $AB = 15$ cm. Find OB.

2. The tangent at P to a circle with centre O cuts a line through O at Q, with $PQ = 24$ cm, $OQ = 25$ cm. Find the radius.

Answers: 1. 17 cm. 2. 7 cm.

Worked Board Problems (Tangent-Length Applications)

1. Find the length of the tangent from a point 25 cm from the centre of a circle of radius 7 cm.

2. ABC is isosceles with $AB = AC$, circumscribed about a circle; prove the base is bisected at the point of contact. *[CBSE 2008, 2012, 2014]*

3. XP, XQ are tangents from X to a circle; R is a point on the circle. Prove $XA+AR = XB+BR$ (with A, B as tangent points from intermediate points). *[CBSE 2014]*

4. Two circles touch externally at C; prove the common tangent at C bisects the other common tangent (between the two touch points). *[CBSE 2013]*

5. AB, BC, CD, DA are tangents to a circle from the vertices of quadrilateral $ABCD$; prove $AB + CD = BC + DA$. *[NCERT; CBSE 2008–2017]*

6. If from an external point B of a circle with centre O, tangents BC, BD are drawn with $\angle DBC = 120°$, prove $BC + BD = BO$, i.e. $BO = 2BC$. *[NCERT Exemplar; HOTS]*

7. A circle touches side BC of $\triangle ABC$ at P and extended sides AB, AC at Q, R; prove $AQ = \frac{1}{2}$(perimeter of $\triangle ABC$). *[CBSE 2000–2023; NCERT Exemplar]*

8. If two tangents PA, PB from an external point P touch a circle at A, B, and a third tangent meets PA, PB at C, D touching the circle at Q, with $PA = 10$ cm, find the perimeter of $\triangle PCD$. *[NCERT Exemplar]*

9. In a hexagon $ABCDEF$ circumscribing a circle, prove $AB + CD + EF = BC + DE + FA$. *[NCERT Exemplar]*

10. If s is the semi-perimeter of $\triangle ABC$ and the incircle touches BC, CA, AB at D, E, F, prove $BD = s - b$. *[NCERT Exemplar]*

11. A chord PQ of a circle is parallel to the tangent at a point R; prove that R bisects the arc PRQ. *[NCERT Exemplar; HOTS]*

12. Prove that the tangent drawn at the mid-point of an arc of a circle is parallel to the chord joining the arc's end-points. *[NCERT Exemplar; HOTS]*

13. Find the locus of the centres of circles which touch a given line at a given point. *[HOTS]*

14. Circles $C(O, r)$ and $C(O', r/2)$ touch internally at A; AB is a chord of $C(O, r)$ meeting $C(O', r/2)$ at C. Prove $AC = CB$. *[HOTS]*

Answers: Tangent length example: 24 cm.

Exercise 8.2

1. PT tangent at T; $OP = 17$ cm, $OT = 8$ cm. Find PT.

2. P is 26 cm from centre O; tangent length $PT = 10$ cm. Find the radius.

3. PA, PB tangents from P; CD another tangent touching at Q. If $PA = 12$ cm, $QC = QD = 3$ cm, find $PC + PD$. *[CBSE 2017]*

4. AB, AC, PQ are tangents (as per figure) with $AB = 5$ cm; find the perimeter of $\triangle APQ$. *[CBSE 2000]*

5. A circle touches all four sides of quadrilateral $ABCD$; $AB = 6, BC = 7, CD = 4$; find AD. *[CBSE 2002]*

6. Two concentric circles of radii 5 cm, 3 cm; from external P, tangent to larger $AP = 12$ cm; find tangent length BP to smaller. *[CBSE 2010, 2012, 2016]*

7. A circle inscribed in quadrilateral $ABCD$ with $\angle B = 90°$; $AD = 23, AB = 29, DS = 5$; find the radius. *[CBSE 2023]*

8. AB is a chord of length 16 cm of a circle of radius 10 cm; tangents at A, B meet at P. Find PA. *[CBSE 2010]*

9. Two concentric circles of diameters 30 cm, 18 cm; find the chord of the larger circle tangent to the smaller. *[CBSE 2014]*

10. $\triangle ABC$ right-angled at B, $BC = 6, AB = 8$; find the radius of the incircle. *[CBSE 2002]*

11. $A\triangle ABC$ circumscribes a circle of radius 4 cm; $BD = 8, DC = 6$; find AB, AC given area $= 84$ cm^2. *[CBSE 2015, 2023]*

12. From external P, $PA = PB$ tangents with $\angle APB = 60°$, $PA = 10$ cm; find chord AB. *[CBSE 2016]*

13. From external P, $PA = PB$ tangents with $\angle PAB = 50°$; find $\angle AOB$. *[CBSE 2016]*

14. Three consecutive sides of a quadrilateral circumscribing a circle are 4, 5, 7 cm; find the fourth side.

15. From an external point P, tangents PA, PB with $OP = 2r$ (r = radius); show $\triangle PAB$ is equilateral. *[CBSE 2008, 2011, 2012]*

16. $\triangle PQR$ circumscribes a circle of radius 8 cm; the point of contact divides QR into segments 14 cm and 16 cm; if area = 336 cm^2, find PQ, PR. *[CBSE 2014]*

17. Equal circles with centres O, O' touch at X; OO' produced meets a circle centred O' at A; AC is tangent to the circle centred O; $O'D \perp AC$. Find $\dfrac{DO'}{CO}$. *[HOTS]*

18. BC is tangent to a circle centred O; OE bisects $\angle AP$ (as per figure). Prove $\triangle AEO \sim \triangle ABC$. *[HOTS]*

19. $PO \perp QO$; tangents at P, Q meet at T. Prove PQ and OT are right bisectors of each other. *[HOTS]*

20. O is the centre of a circle; BCD is tangent to it at C. Prove $\angle BAC + \angle ACD = 90°$. *[CBSE 2023; HOTS]*

Answers: 1. 15 cm. 2. 24 cm. 3. 18 cm. 4. 10 cm. 5. 3 cm. 6. $4\sqrt{10}$ cm. 7. 11 cm. 8. $\frac{40}{3}$ cm. 9. 24 cm. 10. 2 cm. 11. 15 cm, 13 cm. 12. 10 cm. 13. 100°. 14. 6 cm. 16. 26 cm, 28 cm.

Very Short Answer Questions (VSAQs)

1. How many tangents can be drawn to a circle from a point inside it? On it? Outside it?

2. The point of contact of a tangent and a circle is called ________.

3. The angle between the tangent at a point on a circle and the radius through that point is ________.

Answers: 1. 0; 1; 2. 2. Point of contact. 3. 90°.

CHAPTER 9

Trigonometric Ratios

No topics have been removed from Introduction to Trigonometry in the CBSE 2026–27 syllabus. Note: this chapter contains no separately-labelled HOTS section in the source book (its harder problems are folded into the main exercise); the LOTS-tier and CBSE/NCERT-tagged items below represent its most challenging content.

Key Concepts

- For $\triangle ABC$ right-angled at B: $\sin A = \frac{BC}{AC}$, $\cos A = \frac{AB}{AC}$, $\tan A = \frac{BC}{AB}$, and reciprocals $\operatorname{cosec} A, \sec A, \cot A$.
- Standard values at $0°, 30°, 45°, 60°, 90°$ (see table in Chapter 10 for full reference).

Exercise 9.1

1. Given one trig ratio, find the other five: (i) $\sin A = \frac{3}{5}$ (ii) $\cos A = \frac{4}{5}$ (iii) $\tan\theta = \frac{11}{60}$ (iv) $\cot\theta = \frac{12}{35}$.

2. In $\triangle ABC$ right-angled at B, $AB = 24, BC = 7$: find (i) $\sin A, \cos A$ (ii) $\sin C, \cos C$. *[NCERT]*

3. Given $15\cot A = 8$, find $\sin A$ and $\sec A$. *[NCERT]*

4. Given $\cot\theta = \frac{7}{8}$, evaluate (i) $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$ (ii) $\cot^2\theta$. *[NCERT]*

5. If $3\cot A = 4$, check whether $\frac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$ or not. *[NCERT]*

6. If $\angle A, \angle B$ are acute angles with $\cos A = \cos B$, show $\angle A = \angle B$. *[NCERT]*

7. In $\triangle ABC$ right-angled at A, if $\tan C = \sqrt{3}$, find $\sin B\cos C + \cos B\sin C$. *[CBSE 2008]*

8. State True/False with reasons: (i) $\tan A$ is always < 1 (ii) $\sec A = \frac{12}{5}$ is possible for some A (iii) $\cos A$ abbreviates cosecant of A (iv) $\cot A$ is the product of cot and A (v) $\sin\theta = \frac{4}{3}$ is possible for some θ.

Answers: 1. (i) $\cos A = \frac{4}{5}, \tan A = \frac{3}{4}, \ldots$ (ii) $\sin A = \frac{3}{5}, \tan A = \frac{3}{4}, \ldots$. 2. (i) $\sin A = \frac{7}{25}, \cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25}, \cos C = \frac{7}{25}$. 3. $\sin A = \frac{15}{17}, \sec A = \frac{17}{8}$. 7. 1. 8. (i) False (ii) True (iii) False (iv) False (v) False.

Exercise 9.2 — Evaluating Trigonometric Expressions

1. Evaluate: $\sin 45^\circ \sin 30^\circ + \cos 45^\circ \cos 30^\circ$.

2. Evaluate: $\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$.

3. Prove that $(\sqrt{3}+1)(3-\cot 30^\circ) = \tan^3 60^\circ - 2\sin 60^\circ$. *[NCERT Exemplar]*

4. If $\tan(A-B) = \frac{1}{\sqrt{3}}$ and $\tan(A+B) = \sqrt{3}$, $0^\circ < A+B \le 90^\circ$, $A > B$, find A, B. *[NCERT]*

5. If $\sin(A-B) = \frac{1}{2}$ and $\cos(A+B) = \frac{1}{2}$, $0^\circ < A+B \le 90^\circ$, $A > B$, find A, B. *[NCERT]*

6. If $4\cot^2 45^\circ - \sec^2 60^\circ + \sin^2 60^\circ + p = \frac{3}{4}$, find p. *[CBSE 2023]*

7. In $\triangle PQR$, right-angled at Q, $PQ = 3$ cm, $PR = 6$ cm; determine $\angle P, \angle R$. *[NCERT]*

8. Evaluate $2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ$. *[CBSE 2024]*

9. If $A = 60^\circ, B = 30^\circ$, verify $\sin(A+B) = \sin A \cos B + \cos A \sin B$. *[CBSE 2024]*

10. If $\sin(A-B) = \frac{1}{2}, \cos(A+B) = \frac{1}{2}$, $0^\circ < A+B \le 90^\circ, A > B$, find $\angle A, \angle B$. *[CBSE 2024]*

Answers: 1. $\frac{\sqrt{2}(\sqrt{3}+1)}{4}$. 2. $\frac{\sqrt{2}(\sqrt{3}-1)}{4}$. 4. $A = 45^\circ, B = 15^\circ$. 5. $A = 45^\circ, B = 15^\circ$. 7. $\angle P = 60^\circ, \angle R = 30^\circ$. 10. $A = 45^\circ, B = 15^\circ$.

Very Short Answer Questions (VSAQs)

1. Evaluate $\sin^2 60^\circ + 2\tan 45^\circ - \cos^2 30^\circ$. *[CBSE 2019]*

2. If $\tan\theta = 1$, find θ.

3. Evaluate $2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ$. *[CBSE 2024]*

Answers: 1. 0. 2. 45°.

CHAPTER 10

Trigonometric Identities

This edition's Trigonometric Identities chapter covers only the Pythagorean identities ($\sin^2\theta + \cos^2\theta = 1$ and its variants) and does not separately treat trigonometric ratios of complementary angles as a distinct topic here (that content, where it appears in the syllabus, concerns identities of the form $\sin(90^\circ - A) = \cos A$ and has been removed for 2026–27). No items in this chapter required exclusion on that basis.

Key Concepts

- $\sin^2\theta + \cos^2\theta = 1; \quad 1 + \tan^2\theta = \sec^2\theta; \quad 1 + \cot^2\theta = \operatorname{cosec}^2\theta.$

Exercise 10.1 — Proving Identities

Prove the following (a representative board-relevant selection):

1. $(1 - \cos^2 A)\operatorname{cosec}^2 A = 1.$

2. $\dfrac{\cos\theta}{1+\sin\theta} = \dfrac{1-\sin\theta}{\cos\theta}.$

3. $\dfrac{\sin\theta}{1-\cos\theta} = \operatorname{cosec}\theta + \cot\theta.$

4. $\dfrac{1-\sin\theta}{1+\sin\theta} = (\sec\theta - \tan\theta)^2.$

5. $\dfrac{1+\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{1+\sin\theta} = 2\sec\theta.$ *[NCERT]*

6. $\dfrac{1+\sec\theta}{\sec\theta} = \dfrac{\sin^2\theta}{1-\cos\theta}.$ *[NCERT; CBSE 2023]*

7. $\dfrac{\sec A - \tan A}{\sec A + \tan A} = \dfrac{\cos^2 A}{(1+\sin A)^2}.$

8. (i) $\dfrac{1+\sin A}{1-\sin A} = \sec A + \tan A$ *[NCERT]* (ii) $\sqrt{\dfrac{1-\cos A}{1+\cos A}} + \sqrt{\dfrac{1+\cos A}{1-\cos A}} = 2\operatorname{cosec} A.$

9. $\dfrac{\sec\theta - 1}{\sec\theta + 1} + \dfrac{\sec\theta + 1}{\sec\theta - 1} = 2\operatorname{cosec}\theta.$ *[CBSE 2001, 2006C, 2023]*

10. $\dfrac{\tan^2 A}{1+\tan^2 A} + \dfrac{\cot^2 A}{1+\cot^2 A} = 1.$ *[NCERT; CBSE 2008]*

11. $\tan^2 A + \cot^2 A = \sec^2 A \operatorname{cosec}^2 A - 2.$

12. $\dfrac{\tan A}{1+\sec A} - \dfrac{\tan A}{1-\sec A} = 2\operatorname{cosec} A.$ *[NCERT Exemplar]*

13. $\sec A(1-\sin A)(\sec A + \tan A) = 1.$ *[CBSE 2023]*

14. If $x = a\sec\theta + b\tan\theta$, $y = a\tan\theta + b\sec\theta$, prove $x^2 - y^2 = a^2 - b^2$. *[CBSE 2001, 2002C]*

15. If $x\cos\theta + y\sin\theta = a$ and $x\sin\theta - y\cos\theta = b$, prove $a^2 + b^2 = x^2 + y^2$. *[CBSE 2024]*

16. If $\operatorname{cosec}\theta + \cot\theta = m$ and $\operatorname{cosec}\theta - \cot\theta = n$, prove $mn = 1$.

17. $(\sin\theta + \cos\theta)(\tan\theta + \cot\theta) = \sec\theta + \operatorname{cosec}\theta.$ *[NCERT Exemplar]*

18. $\dfrac{1+\sec\theta-\tan\theta}{1+\sec\theta+\tan\theta} = \dfrac{1-\sin\theta}{\cos\theta}.$ *[NCERT Exemplar; CBSE 2024]*

19. $\dfrac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta} = \dfrac{1-\cos\theta}{1+\cos\theta}.$ *[NCERT Exemplar]*

20. If $3\sin\theta + 5\cos\theta = 5$, prove $5\sin\theta - 3\cos\theta = \pm 3$.

21. If $\sin\theta + 2\cos\theta = 1$, prove $2\sin\theta - \cos\theta = 2$. *[NCERT Exemplar]*

22. If $\cos A + \cos^2 A = 1$, prove $\sin^2 A + \sin^4 A = 1$. *[CBSE 2023; HOTS]*

23. If $\sin\theta + \cos\theta = k$, prove $\sin^6\theta + \cos^6\theta = 1 - \dfrac{3(k^2-1)^2}{4}$. *[HOTS]*

24. If $x = a\cos^3\theta,\ y = b\sin^3\theta$, prove $\left(\dfrac{x}{a}\right)^{2/3} + \left(\dfrac{y}{b}\right)^{2/3} = 1$. *[HOTS]*

Exercise 10.2 — Evaluating Given a Trig Ratio

1. If $\sin\theta = \dfrac{1}{\sqrt{2}}$, find all other trig ratios of θ.

2. If $\tan\theta = \dfrac{1}{\sqrt{2}}$, find $\dfrac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \cot^2\theta}$.

3. If $4\tan\theta = 3$, evaluate $\dfrac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}$. *[CBSE 2018]*

4. If $\tan\theta = \dfrac{12}{5}$, find $\dfrac{1+\sin\theta}{1-\sin\theta}$.

5. If $\cot\theta = \sqrt{3}$, find $\dfrac{\operatorname{cosec}^2\theta + \cot^2\theta}{\operatorname{cosec}^2\theta - \sec^2\theta}$.

6. If $3\cos\theta = 1$, find $\dfrac{6\sin^2\theta + \tan^2\theta}{4\cos\theta}$.

7. If $\sqrt{3}\tan\theta = 3\sin\theta$, find $\sin^2\theta - \cos^2\theta$. *[CBSE 2001]*

8. If $\sin\theta + \cos\theta = \sqrt{2}\sin\theta$, find $\cot\theta$.

9. If $2\sin^2\theta - \cos^2\theta = 2$, find θ. *[NCERT Exemplar]*

10. If $\sqrt{3}\tan\theta - 1 = 0$, find $\sin^2\theta - \cos^2\theta$. *[NCERT Exemplar]*

11. If $\dfrac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$, find θ. *[HOTS]*

12. If $\cos\theta + \cos^2\theta = 1$, find $\sin^2\theta + \sin^4\theta$. *[HOTS]*

13. If $\dfrac{\tan^4\theta - 1}{\tan^2\theta - 1} = A\sec^2\theta + B\tan\theta$, find $A + B$. *[HOTS]*

14. If $\sin^4 A - \cos^4 A = 1$ and $0 < A < 90°$, find A. *[HOTS]*

Answers: 1. $\cos\theta = \frac{1}{\sqrt{2}}, \tan\theta = 1, \sec\theta = \sqrt{2}, \operatorname{cosec}\theta = \sqrt{2}, \cot\theta = 1$. 3. $\frac{13}{11}$. 4. 25. 6. $\frac{21}{10}$. 7. $\frac{1}{3}$. 8. $\sqrt{2} - 1$. 9. 90°. 10. $\frac{1}{2}$. 11. 90°. 12. 1. 13. $A = 1, B = 0$, so $A + B = 1$. 14. 90°.

Very Short Answer Questions (VSAQs)

1. Evaluate $(1 + \tan^2\theta)(1 - \sin\theta)(1 + \sin\theta)$. *[CBSE 2020]*

2. If $\operatorname{cosec}\theta = 2x$ and $\cot\theta = \dfrac{2}{x}$, find $2\left(x^2 - \dfrac{1}{x^2}\right)$. *[CBSE 2010]*

3. The value of $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$ is ________. *[HOTS]*

Answers: 1. 1. 2. $\frac{1}{2}$. 3. 1.

Fill in the Blanks

1. $\dfrac{\sin^4\theta - \cos^4\theta}{\sin^2\theta - \cos^2\theta} =$ ________.

2. If $\dfrac{1}{1 + \sin\theta} + \dfrac{1}{1 - \sin\theta} = k\sec^2\theta$, then $k =$ ________.

3. If $\sin\theta - \cos\theta = \dfrac{3}{5}$, then $\sin\theta\cos\theta =$ ________.

4. If $\cos\theta + \cos^2\theta = 1$, then $\sin^2\theta + \sin^4\theta =$ ________.

5. If $\sin^4 A - \cos^4 A = 1$ and $0 < A < 90°$, then $A =$ ________.

Answers: 1. 1. 2. 2. 3. $\frac{8}{25}$. 4. 1. 5. 90°.

CHAPTER 11

Heights and Distances

Per the CBSE 2026–27 syllabus, numerical problems are restricted to angles of $30^\circ, 45^\circ, 60^\circ$ *and to at most two right triangles; one item using non-standard angles* ($32^\circ, 63^\circ$) *has been excluded. Problems whose premise is that two angles of elevation are* ***complementary*** *(testing the complementary-ratio identity* $\tan\theta\tan(90^\circ-\theta)=1$*) have also been excluded, consistent with the removal of trigonometric ratios of complementary angles.*

Key Concepts

- Angle of elevation/depression measured from the horizontal.
- Set up right triangle(s) using the given angle(s) and known length(s); solve with $\tan, \sin, \cos$ as appropriate.

Exercise 11.1

1. A tower stands vertically on the ground; from a point 20 m from the foot, the angle of elevation of the top is 60°. Find the height.

2. A ladder leaning against a wall makes 60° with the ground; the foot is 9.5 m from the wall. Find the ladder's length.

3. A 15 m ladder just reaches the top of a wall, making 60° with the wall. Find the wall's height. *[NCERT Exemplar]*

4. A tower is surmounted by a flagstaff; from a point 70 m away, the angles of elevation of the top and bottom of the flagstaff are $60^\circ, 45^\circ$. Find the flagstaff's and tower's heights. *[CBSE 2014]*

5. Two points A, B are on the same side of a tower in line with its base; angles of depression from the top are $60^\circ, 45^\circ$. If the tower is 15 m, find AB. *[CBSE 2017]*

6. A tower is surmounted by a 5 m flagstaff; angles of elevation of bottom and top of the flagstaff are $30^\circ, 60^\circ$. Find the tower's height. *[CBSE 2015, 2016, 2019, 2020]*

7. Elevation of a tower is 30°; advancing 150 m, it becomes 60°. Show the height is 129.9 m (use $\sqrt{3}=1.732$). *[CBSE 2006]*

8. Elevation of a tower's top from point A is 30°; moving 20 m to B, it becomes 60°. Find the height and distance from A. *[CBSE 2002, 2013, 2017]*

9. From the top of a 15 m building, elevation of a tower's top is 30°; from the building's base, it's 60°. Find the tower's height and distance from the building. *[CBSE 2002]*

10. A tower with a flagpole; at a point 9 m from the foot, elevations of top/bottom of the flagpole are $60°, 30°$. Find both heights. *[CBSE 2005]*

11. A 1.5 m tall boy stands some distance from a 30 m building; elevation increases from $30°$ to $60°$ as he walks towards it. Find the distance walked. *[NCERT; CBSE 2014]*

12. A tower's shadow is 40 m longer when the sun's altitude is $30°$ than at $60°$. Find the tower's height. *[NCERT Exemplar; CBSE 2019]*

13. From the ground, elevations of the bottom and top of a transmission tower on a 20 m building are $45°, 60°$. Find the transmission tower's height. *[NCERT; CBSE 2023]*

14. Angles of depression of the top and bottom of an 8 m building from a multistoried building are $30°, 45°$. Find the multistoried building's height and the distance between them. *[NCERT; CBSE 2009]*

15. A 1.6 m statue stands on a pedestal; from a point on the ground, elevation of the statue's top is $60°$, of the pedestal's top is $45°$. Find the pedestal's height. *[NCERT; CBSE 2008, 2014]*

16. From a 120 m tower, a man observes two cars on opposite sides with depression angles $60°, 45°$. Find the distance between the cars (use $\sqrt{3} = 1.732$). *[CBSE 2017]*

17. From a 7 m building, elevation of a cable tower's top is $60°$, depression of its foot is $45°$. Find the tower's height. *[NCERT; CBSE 2014, 2017, 2023]*

18. From a 75 m lighthouse, depression angles of two ships (in line) are $30°, 45°$. Find the distance between them. *[NCERT; CBSE 2013, 2023]*

19. Elevation of a building's top from a tower's foot is $30°$; elevation of the tower's top from the building's foot is $60°$. If the tower is 50 m, find the building's height. *[NCERT; CBSE 2012, 2017]*

20. From a bridge 30 m above a river, depression angles of the banks are $30°, 45°$. Find the river's width. *[NCERT]*

21. Elevation of a hill's top from a tower's foot is $60°$; elevation of the tower's top from the hill's foot is $30°$. If the tower is 50 m, find the hill's height. *[CBSE 2006C, 2013]*

22. (i) Two boats approach a lighthouse from opposite directions; elevations $30°, 45°$; distance between boats 100 m. Find the lighthouse's height. *[CBSE 2014]* (ii) From a 45 m lighthouse, depression of two ships (opposite sides) are $30°, 60°$. Find the distance between them (use $\sqrt{3} = 1.73$). *[CBSE 2024]*

23. Two men on either side of an 80 m cliff observe elevations $30°, 60°$. Find the distance between them. *[CBSE 2016]*

24. A flagstaff on a 5 m tower; elevation of the flagstaff's top is $60°$, of the tower's top is $45°$. Find the flagstaff's height. *[CBSE 2013]*

25. Horizontal distance between two poles is 15 m; depression of the first pole's top from the second (24 m high) is $30°$. Find the first pole's height (use $\sqrt{3} = 1.732$). *[CBSE 2013]*

26. Depression angles of two ships from a lighthouse (same side) are $45°, 30°$; ships are 200 m apart. Find the lighthouse's height. *[CBSE 2012]*

27. From a bridge 8 m above the banks, depression angles are $30°, 45°$. Find the river's width. *[CBSE 2022]*

28. Two stations due south of a leaning tower (leaning towards the north) are at distances a, b from its foot. If α, β are the elevations of the top from these stations, prove the tower's inclination θ to the horizontal satisfies $\cot\theta = \dfrac{b\cot\alpha - a\cot\beta}{b-a}$. *[HOTS]*

Answers: 1. $20\sqrt{3}$ m. 3. 7.5 m. 5. $15(\sqrt{3}-1)$ m. 6. 2.5 m. 12. $20\sqrt{3}$ m. 15. $20(\sqrt{3}-1) \approx 14.6$ m. 19. $7(\sqrt{3}+1)$ m. 20. Height $= 25\sqrt{3}$ m. 27. Proved (general relation).

Exercise 11.1 (continued) — LOTS

1. From a 60 m building, depression angles of the top and bottom of a lamp post are $30°, 60°$. Find (i) the horizontal distance (ii) the lamp post's height (iii) the height difference (iv) the distance between the tops. *[CBSE 2009, 2024]*

2. A boat's depression angle from a 150 m cliff changes from $60°$ to $45°$ in 2 minutes. Find its speed in m/h. *[CBSE 2017]*

3. A boat rowing away from a 100 m lighthouse takes 2 minutes for the elevation to change from $60°$ to $30°$. Find the boat's speed (use $\sqrt{3} = 1.732$). *[CBSE 2019]*

CHAPTER 12

Areas Related to Circles

No topics have been removed from Areas Related to Circles. Per the syllabus, segment-area problems are restricted to central angles of $60°, 90°$ *or* $120°$*; all segment problems below satisfy this.*

Key Concepts

- Circumference $= 2\pi r$; area $= \pi r^2$.
- Arc length $= \dfrac{\theta}{360} \times 2\pi r$; sector area $= \dfrac{\theta}{360} \times \pi r^2$.
- Segment area = sector area − triangle area (minor); major segment/sector $= \pi r^2 -$ corresponding minor.

Exercise 12.1 — Circumference & Area of a Circle

1. Find the circumference and area of a circle of radius 4.2 cm.

2. A road 7 m wide surrounds a circular park of circumference 352 m. Find the area of the road.

3. The circumferences of two circles are in ratio $2 : 3$. Find the ratio of their areas.

4. The sum of the radii of two circles is 140 cm and the difference of their circumferences is 88 cm. Find the diameters. *[NCERT]*

5. Find the circumference of a circle whose area equals the sum of the areas of two circles of radii 15 cm, 18 cm. *[NCERT]*

6. A path 21 m wide surrounds a circular garden; the area of the path is a given value — find the garden's radius. *[NCERT Exemplar]*

7. Find the area between two concentric circles inscribed and circumscribing a square (of given side). *[NCERT Exemplar]*

8. The area of a circle inscribed in an equilateral triangle is 154 cm^2; find the triangle's perimeter. *[NCERT Exemplar]*

9. A steel wire bent into a square encloses an area of 121 cm^2; the same wire is bent into a circle. Find the circle's area. *[NCERT Exemplar]*

10. The wheel of a vehicle has a given diameter; find the number of revolutions to cover a stated distance. *[CBSE 2019, 2023]*

11. Find the area of a sector of a circle of radius 21 cm subtending an angle of 60° at the centre. *[NCERT Exemplar]*

12. A car has wheels of diameter 80 cm; find the number of complete revolutions in 10 minutes at 66 km/h. *[CBSE 2024]*

13. Find the area enclosed between two concentric circles of radii 3.5 cm and 7 cm. A third concentric circle is drawn outside the 7 cm circle such that the area between it and the 7 cm circle equals that between the two inner circles. Find the radius of the third circle, correct to one decimal place. *[HOTS]*

Answers: 1. Circumference = 26.4 cm, area = 55.44 cm^2. 2. Area of road = 2926 m^2 (approx.). 3. 4 : 9. 13. Radius ≈ 8.6 cm.

Exercise 12.2 — Arc Length & Sector Area

1. Find, in terms of π, the arc length subtending 30° at the centre of a circle of radius 4 cm.

2. Find the angle subtended at the centre by an arc of length $\dfrac{5\pi}{3}$ cm in a circle of radius 5 cm.

3. An arc of length 20π cm subtends 144° at the centre; find the radius.

4. A car's wheel wipers sweep a given angle; find the area cleaned at each sweep. *[CBSE 2013, 2017, 2023]*

5. Find the area swept by the minute hand of a clock of given length between 6:05 and 6:40 a.m. *[NCERT Exemplar]*

6. A round table cover has six equal designs; find the cost of making the designs at a given rate. *[NCERT, CBSE 2019, 2023]*

7. Find the area of a sector whose arc subtends a given angle, and the corresponding segment area. *[CBSE 2024]*

8. A chord of a circle subtends an angle of 30° (or similar) at the centre; find the radius given the arc length. *[NCERT Exemplar]*

9. In a circle of given radius, $\angle AOB = 30°$; find the area of the shaded (segment/sector) region. *[NCERT; CBSE 2012]*

10. A bicycle wheel makes a given number of revolutions per minute; find its speed in km/h. *[CBSE 2024]*

11. An archery target has three regions formed by three concentric circles with diameters in ratio 1 : 2 : 3. Find the ratio of the areas of the three regions. *[NCERT Exemplar; HOTS]*

12. A chord subtends an angle θ at the centre; the minor segment cut off is $\dfrac{1}{8}$ of the circle's area. Prove $8\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2} + \pi = \dfrac{\pi\theta}{45}$. *[HOTS]*

Answers: 1. $\dfrac{2\pi}{3}$ cm. 2. 60°. 3. 25 cm. 11. Ratio = 1 : 3 : 5. 12. Proved.

Exercise 12.3 — Segment Area

1. A chord PQ of length 12 cm subtends $120°$ at the centre. Find the area of the minor segment. *[NCERT]*

2. A chord AB of a circle of radius 14 cm subtends $60°$ at the centre. Find the areas of the minor and major segments. *[NCERT Exemplar; CBSE 2019, 2023]*

3. A chord of a circle of radius 20 cm subtends $90°$ at the centre. Find the area of the major segment (use $\pi = 3.14$). *[NCERT Exemplar]*

4. A chord 10 cm long is drawn in a circle of radius $5\sqrt{2}$ cm. Find the areas of both segments (use $\pi = 3.14$).

5. Radii OA, OB of a circle of radius 5 cm are at right angles; find the areas of both segments cut off by chord AB (use $\pi = 3.14$).

Answers: 1. $4(4\pi - 3\sqrt{3})$ cm^2. 2. 17.80 cm^2, 588.20 cm^2. 3. 1142 cm^2. 4. 14.25 cm^2, 142.75 cm^2. 5. 7.135 cm^2, 71.425 cm^2.

Very Short Answer Questions (VSAQs)

1. If the diameter of a semicircular protractor is 14 cm, find its perimeter. *[CBSE 2009]*

2. Find the radius of a circle given the arc length and angle subtended at the centre. *[CBSE 2020]*

3. Find the perimeter of a sector of a circle of radius 5.2 cm given its arc length.

4. Find the area of a quadrant of a circle of radius 42 cm. *[CBSE 2024]*

5. A square is inscribed in a quadrant of a circle; find the area of the shaded corners (use $\pi = 3.14$). *[CBSE 2024]*

CHAPTER 13

Surface Areas and Volumes

*Per the CBSE 2026–27 syllabus, the **frustum of a cone** has been removed. This edition's exercises are already focused on combination solids and conversion-of-solids problems (no frustum computation appears in the main exercise or VSAQs); two Fill-in-the-Blank shape-identification items naming frustum-shaped objects (funnel, shuttlecock) have been excluded below.*

Key Concepts

- Surface area/volume of a combination of any two of: cuboid, cone, cylinder, sphere, hemisphere — add curved surfaces only (not internal joining faces).
- Conversion of solids: volume is conserved when one shape is recast into another.

Exercise 13.1

1. A toy is a cylinder (radius 5 cm, height 13 cm) with a hemisphere on one end and a cone on the other, same radius, total height 30 cm. Find its surface area. *[CBSE 2002]*

2. A cylindrical tub (radius 5 cm, length 9.8 cm) is full of water; a cone-on-hemisphere solid (hemisphere radius 3.5 cm, cone height outside 5 cm) is immersed. Find the water left. *[CBSE 2000C]*

3. A solid is a cylinder with hemispherical ends, total length 104 cm, hemisphere radius 7 cm. Find the cost of polishing at Rs 10 per dm^2. *[CBSE 2006C]*

4. A vessel is a hollow hemisphere (diameter 14 cm) mounted by a hollow cylinder, total height 13 cm. Find the inner surface area. *[CBSE 2013]*

5. A toy is a cone (radius 3.5 cm) on a hemisphere of the same radius, total height 15.5 cm. Find the total surface area. *[CBSE 2013]*

6. A cylindrical vessel (diameter 10 cm, height 10.5 cm) full of water has a cone (diameter 7 cm, height 6 cm) immersed. Find the water displaced and left. *[CBSE 2009]*

7. A hemispherical depression (diameter = edge) is cut from a 21 cm cubical block. Find the volume and total surface area of the remainder. *[CBSE 2010]*

8. A toy is a hemisphere surmounted by a cone of the same base radius 21 cm; the cone's volume is $\frac{2}{3}$ of the hemisphere's. Find the cone's height and the toy's surface area. *[CBSE 2010]*

9. A solid is a cone on a hemisphere, both radius 3.5 cm, total height 9.5 cm. Find the volume. *[CBSE 2012]*

10. The largest possible sphere is carved from a 7 cm wooden cube. Find the volume of wood left. *[CBSE 2014]*

11. From a solid cylinder (height 2.8 cm, diameter 4.2 cm), a conical cavity of the same height/diameter is hollowed out. Find the total surface area of the remainder. *[CBSE 2014]*

12. The largest cone is carved from a 21 cm solid cube. Find the volume of the remainder. *[CBSE 2015]*

13. A wooden pen stand is a cuboid ($10 \times 5 \times 4$ cm) with 4 conical depressions (radius 0.5 cm, depth 2.1 cm) and one cubical depression (edge 3 cm). Find the volume of wood. *[NCERT Exemplar]*

14. A solid toy is a hemisphere surmounted by a cone (height 4 cm, base diameter 8 cm). Find the toy's volume, the volume difference with a circumscribing cube, and the toy's surface area. *[NCERT Exemplar]*

15. Marbles of diameter 1.4 cm are dropped into a cylindrical beaker until the water rises by 5.6 cm; find the number of marbles. *[NCERT Exemplar]*

16. Two cubes of volume 64 cm^3 each are joined end to end; find the surface area of the resulting cuboid. *[NCERT Exemplar]*

17. A cylindrical block is used to build a wall; find the number of bricks needed given block/wall dimensions. *[NCERT Exemplar]*

18. A building is a cylinder surmounted by a hemispherical dome; given the dome's diameter is $\frac{2}{3}$ the total height, and it contains $67\frac{1}{21}$ m^3 of air, find the height. *[NCERT Exemplar; CBSE 2023]*

19. Two cones of given dimensions sit inside a cylinder; find the volume of the remaining portion of the cylinder. *[NCERT Exemplar]*

20. An ice-cream cone (radius, height given) is filled with a hemispherical top; find the volume of ice-cream. *[CBSE 2015, 2023]*

21. Water flows through a cylindrical pipe of given radius and speed into a conical vessel of given dimensions; find the time to fill it. *[CBSE 2024]*

22. A right triangle with sides 15 cm and 20 cm is revolved about its hypotenuse. Find the volume and surface area of the double cone so formed (use $\pi = 3.14$). *[HOTS]*

Answers: 9. ≈ 214.5 cm^3. 10. ≈ 129.4 cm^3. 16. 160 cm^2. 22. Volume $= 1200\pi = 3768$ cm^3; surface area $= 420\pi \approx 1318.8$ cm^2 (with $OA = 12$ cm, $OB = 9$ cm, $BC = 25$ cm).

Very Short Answer Questions (VSAQs)

1. Radii of a cylinder and cone are in ratio 3 : 4, heights in ratio 2 : 3. Find the ratio of their volumes.

2. Heights of two cones are in ratio 1 : 2, base perimeters in ratio 3 : 4. Find the ratio of their volumes.

3. A cone and sphere have equal radii and equal volumes. Find the ratio of the sphere's diameter to the cone's height.

4. A cone, hemisphere and cylinder stand on equal bases with the same height. Find the ratio of their volumes.

5. Radii of two cylinders are in ratio 3 : 5, heights in ratio 2 : 3. Find the ratio of their curved surface areas.

6. Two cubes have volumes in ratio 1 : 27. Find the ratio of their surface areas.

7. Two right circular cylinders of equal volume have heights in ratio 1 : 2. Find the ratio of their radii.

8. The surface area of a sphere is 616 cm^2. Find its radius. *[CBSE 2008]*

9. A cylinder and cone have the same base radius and height. Find the ratio of their volumes. *[CBSE 2009]*

10. The volume and surface area of a solid hemisphere are numerically equal. Find its diameter. *[CBSE 2017]*

11. A solid metallic sphere of radius 3 cm is melted and recast as a cylinder of radius 2 cm. Find the height. *[CBSE 2022]*

12. Radii of two cones are in ratio 2 : 1 with equal volumes. Find the ratio of their heights.

13. A cuboid ($11 \times 7 \times 7$ cm) is melted into spheres of radius $\frac{7}{2}$ cm; find the number of spheres. *[CBSE 2022]*

Answers: 1. 9 : 8. 2. 9 : 32. 3. 1 : 2. 4. 1 : 2 : 3. 5. 2 : 5. 6. 1 : 9. 7. $\sqrt{2}$: 1. 8. 7 cm. 9. 3 : 1. 10. 9 units. 11. 9 cm. 12. 1 : 4. 13. 3.

Fill in the Blanks

1. A cylindrical pencil sharpened at one edge is the combination of a _____ and a _____. *[NCERT Exemplar]*

2. A *surahi* is the combination of a _____ and a _____. *[NCERT Exemplar]*

3. A plumbline (*sahul*) is the combination of a _____ and a _____. *[NCERT Exemplar]*

4. The shape of a *gilli*, in the gilli-danda game, is a combination of _____ and a _____. *[NCERT Exemplar]*

5. Two identical solid cubes of side a are joined end to end. The total surface area of the resulting cuboid is ________.

6. If a solid cone of base radius r and height h is placed on a cylinder of the same base radius and height, the curved surface area of the shape is ________. *[NCERT Exemplar]*

7. A solid ball is exactly fitted inside a cuboidal box of side a. The ball's volume is ________.
[NCERT Exemplar]

8. Two identical solid hemispheres of radius r are stuck together along their bases; the total surface area of the combination is ________.

9. If a marble of radius 2.1 cm is put into a cylindrical cup (radius 5 cm, height 6 cm) full of water, the volume of water that flows out is ________. *[NCERT Exemplar]*

10. The volume of the greatest right circular cone cut from a cube of side 4 cm is ________.

Answers: 1. Cone, cylinder. 2. Sphere, cylinder. 3. Hemisphere, cone. 4. Two cones, a cylinder. 5. $10a^2$. 6. $\pi r(\sqrt{r^2+h^2}+2h)$. 7. $\dfrac{1}{6}\pi a^3$. 8. $4\pi r^2$. 9. ≈ 38.8 cm^3. 10. $\dfrac{32\pi}{3}$ cm^3.

CHAPTER 14

Statistics

*Per the CBSE 2026–27 syllabus, the graphical representation of cumulative frequency distributions (**ogives**) has been removed. This edition's exercises use cumulative frequency only in tabular form for computing the mean/median by formula (retained); the one VSAQ asking to define/compare the ogive types has been excluded below. Note: this chapter contains no separately-labelled HOTS section in the source book; its LOTS-tier problems (multi-method mean computations, missing-frequency problems) represent its most challenging content.*

Key Concepts

- Mean: Direct method $\bar{x} = \dfrac{\sum f_i x_i}{\sum f_i}$; Assumed Mean $\bar{x} = a + \dfrac{\sum f_i d_i}{\sum f_i}$; Step-deviation $\bar{x} = a + h\dfrac{\sum f_i u_i}{\sum f_i}$.
- Mode $= l + \left(\dfrac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h$ (modal class values).
- Median $= l + \left(\dfrac{\frac{n}{2} - cf}{f}\right) h$ (median class values, via cumulative frequency table).

Exercise 14.1–14.3 — Mean (Direct, Assumed Mean, Step-Deviation)

1. Calculate the mean: x: 5,6,7,8,9; f: 4,8,14,11,3.

2. Find the missing p if the mean of x: 5,8,10,12,20,25; f: 2,5,8,22,p,4,2 is 12.58.

3. Ages of 40 students: 15–20 years with given frequencies; find the mean age.

4. Five coins tossed 1000 times; number of heads (0–5) with given frequencies. Find the mean number of heads per toss.

5. The mean of x: 5,10,15,20,25 with frequencies 7,k,8,4,5 is 14; find k. *[CBSE 2002C]*

6. The mean of x: 5,15,25,35,45 with frequencies 3,k,3,6,2 is 25; find k. *[CBSE 2001]*

7. A survey of 20 houses recorded the number of plants (grouped 0–14); find the mean number of plants per house and justify the method used. *[NCERT]*

8. Heart-beat data of 30 women (grouped 65–86); find the mean heart rate. *[NCERT]*

9. Daily pocket allowance of children in a building, mean given as Rs 18; find the missing frequency. *[NCERT]*

10. Mean of a grouped distribution is 27; find the missing frequency p. *[CBSE 2006C]*

11. Daily expenditure on food of 25 households (grouped); find the mean by a suitable method. *[NCERT; CBSE 2019]*

12. Marks of 110 students (grouped 30–65); find the mean marks. *[CBSE 2019]*

13. Cumulative frequency distribution (less-than type) of 1000 persons aged 20+; determine the mean age. *[NCERT Exemplar]*

Answers: 1. $\bar{x} \approx 7.15$. 5. $k = 2$. 6. $k = 6$.

Exercise 14.4 — Median

1. Find the median of: 715, 724, 725, 710, 729, 745, 694, 699, 696, 712, 734, 728, 716, 705, 719.

2. Calculate the median salary of 280 persons from grouped salary data (5–50, thousands). *[CBSE 2018]*

3. Find the missing frequency given the median is 24, from grouped age data (0–50).

4. Life-times of 400 neon lamps (grouped); find the median life. *[NCERT]*

5. Policy holders' ages (cumulative data given); find the median age. *[NCERT]*

6. Leaf lengths (grouped, in mm); find the median length. *[NCERT]*

7. Find missing frequencies x, y given the median and total frequency of a grouped distribution. *[NCERT Exemplar]*

8. Weekly expenditure data (grouped); find x, the median, and the mean expenditure. *[CBSE 2023]*

Answers: 1. Median $= 716$.

Exercise 14.5 — Mode

1. Find the mode: (i) $3, 5, 7, 4, 5, 3, 5, 6, 8, 9, 5, 3, 5, 3, 6, 9, 7, 4$ (ii) a modified version of the same set (iii) $15, 8, 26, 25, 24, 15, 18, 20, 24, 15, 19, 15$.

2. Shirt sizes of 200 people (grouped by size); find the mode.

3. Determine the modal lifetimes of components from grouped life-time data. *[NCERT]*

4. Find the mean, mode and median of a given grouped data set and compare them. *[CBSE 2009]*

5. Distribution of states/UTs by a demographic variable (grouped); find the mode. *[NCERT]*

6. Find the modal agricultural holdings of a village from grouped data. *[NCERT Exemplar]*

7. Calculate the modal income from grouped income data. *[NCERT Exemplar]*

Very Short Answer Questions (VSAQs)

1. Find the class marks of the classes 10–25 and 35–55. *[CBSE 2008]*

2. If the mean of a grouped distribution with a missing frequency x is known, find x. *[CBSE 2022]*

CHAPTER 15

Probability

No topics have been removed from Probability in the CBSE 2026–27 syllabus.

Key Concepts

- $P(E) = \dfrac{\text{number of favourable outcomes}}{\text{number of all equally likely outcomes}}$; $0 \le P(E) \le 1$; $P(E) + P(\bar{E}) = 1$.

Exercise 15.1

1. The probability of rain tomorrow is 0.85. Find the probability it will not rain.

2. A die is thrown; find the probability of (i) a prime number *[NCERT; CBSE 2019]* (ii) 2 or 4 (iii) a multiple of 2 or 3 (iv) an even prime number *[CBSE 2008]* (v) a number greater than 5 *[CBSE 2008]* (vi) a number between 2 and 6 (vii) a composite number. *[NCERT; CBSE 2019]*

3. Two unbiased dice are thrown; find the probability the total exceeds 10. *[CBSE 2018]*

4. A box has 20 cards numbered 1–20; find the probability the drawn card is (i) divisible by 2 or 3 (ii) a prime number. *[CBSE 2015]*

5. A bag has balls of different colours; find the probability of drawing (i) white (ii) red (iii) black (iv) not red. *[CBSE 2008]*

6. From a well-shuffled deck, find the probability of drawing (iii) a perfect square (iv) an even prime number. *[CBSE 2014]*

7. A purse has coins of different denominations; find the probability the coin drawn (iii) is worth less than Rs 5 (iv) is a Re 1 or Rs 2 coin. *[CBSE 2014]*

8. Cards numbered from a given range; find the probability the number is (i) not divisible by 3 *[CBSE 2005]* (ii) a prime number greater than 7 (iii) not a perfect square. *[CBSE 2014]*

9. From a deck of cards, find the probability of drawing (i) white (ii) red (iii) not black (iv) red or white *[CBSE 2004]* (vii) queen of hearts *[CBSE 2024]* (viii) not a jack. *[CBSE 2024]*

10. A card is drawn from a deck; find the probability it is (i) an ace (ii) a king. *[CBSE 2014; NCERT]*

11. Cards from a given range; find a stated probability. *[CBSE 2013]*

12. A number is selected from a given range, each equally likely; find a stated probability. *[CBSE 2005]*

13. Cards of different colours; find the probability of neither yellow nor blue. *[CBSE 2012]*

14. Find the probability of (i) a prime number (ii) a multiple of 7 (iii) a multiple of 3 or 5. *[CBSE 2006C]*

15. Find the probability of a number between 15 and a multiple of 5. *[CBSE 2008]*

16. The probability two students don't share a birthday is 0.992; find the probability they do. *[NCERT]*

17. Find the probability of (i) divisible by 5 (ii) a perfect square. *[CBSE 2007]*

18. Two coins are tossed; find the probability of at least one head. *[NCERT]*

19. Three friends' birthdays: find the probability of (i) same day (ii) different days (iii) consecutive days. *[NCERT]*

20. Two customers visit a shop in the same week (Monday–Saturday), each equally likely on any day. Find the probability both visit on (i) the same day (ii) different days (iii) consecutive days. *[NCERT; HOTS]*

Answers: 1. 0.15. 2. (i) $\frac{1}{2}$ (ii) $\frac{1}{3}$ (iii) $\frac{2}{3}$ (iv) $\frac{1}{6}$ (v) $\frac{1}{6}$ (vi) $\frac{1}{2}$ (vii) $\frac{1}{3}$. 3. $\frac{1}{12}$. 10. (i) $\frac{1}{13}$ (ii) $\frac{1}{13}$. 16. 0.008. 18. $\frac{3}{4}$. 19. (i) $\frac{1}{6}$ (ii) $\frac{5}{6}$ (iii) $\frac{5}{18}$.

Very Short Answer Questions (VSAQs)

1. A die is thrown once; find the probability of getting a number greater than 4. *[CBSE 2010]*

2. A die is thrown once; find the probability of getting a multiple of 4. *[CBSE 2010]*

3. A letter of the English alphabet is chosen; find the probability it is a consonant. *[CBSE 2015, 2020]*

4. A bag has balls of various colours; find the probability the drawn ball is not red. *[CBSE 2017]*

5. A number is chosen from a given range; find the probability its square is ≤ 1. *[CBSE 2017]*

6. If the probability of winning a game is 0.07, find the probability of losing. *[CBSE 2020]*

Answers: 1. $\frac{1}{3}$. 2. $\frac{1}{6}$. 6. 0.93.